MASTER'S DEGREE IN PHARMACEUTICAL BIOTECHNOLOGY

From Genome to Drug – Exploring Bioinformatics & Omics Sciences

Lesson III - Transcriptomics Analysis

Transcriptomics – DESEq2 and Visualization

Estevão Barcelos, PostDoc UNIPG

estevaocarlos.silvabarcelos@collaboratori.unipg.it

Learning Objectives — Lesson III

Introduction to Transcriptomics

Cover gene expression, RNA sequencing (RNA-seq), and the biological questions transcriptomics can answer.

Transcriptomic Datasets

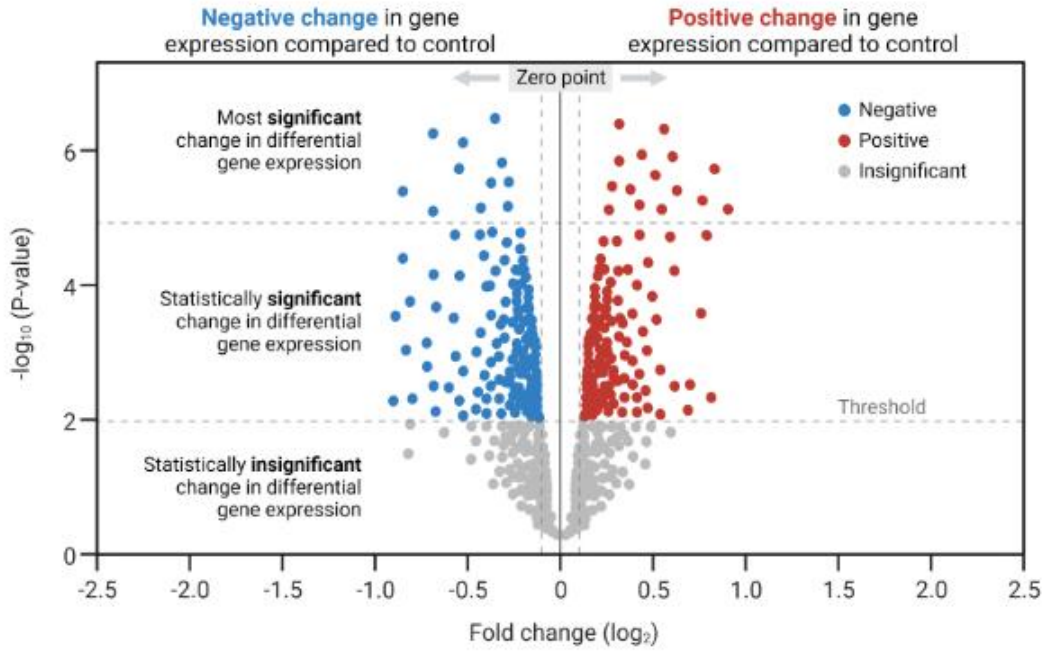
Introduce count matrices, sample metadata, biological replicates, and public datasets such as GEO or ArrayExpress.

Differential Gene Expression: DESeq2

Cover experimental design, normalization, statistical testing, adjusted p-values, and $\log_2$ fold changes.

Data Visualisation: Volcano Plots

Explain $\log_2$ fold change, statistical significance, upregulated and downregulated genes, and selection thresholds.



QUIZ

CASE 1

A BLASTp search returns a hit with an *E-value* = 3.2.

What is the most appropriate conclusion?

Hint: The E-value tells you how many hits this good you would expect by pure chance in a database of this size.

CASE 2

You BLAST an unknown protein sequence and get a hit with *E-value* = 0.0, 100% query coverage, and 76% identity.

What can you confidently conclude?

CASE 3

A somatic mutation in a cancer patient changes the codon CAA → UAA in the *CTLA-4* gene.

What type of mutation is this, and what is the likely consequence?

Noise

An E-value of 3.2 means you would expect ~3 hits this good purely by random chance. This is not statistically significant — it is almost certainly noise. E-values well below 0.05 (ideally < 1e-5) are needed to consider a hit meaningful, and E < 1e-10 indicates a very strong true homolog.

Homolog

The hit is a very strong homolog: the sequences are nearly identical across their full length, almost certainly sharing the same biological function.

Nonsense mutation

CAA encodes Glutamine (Gln). UAA is the ochre stop codon. This single base change (C→U) is a nonsense mutation — the ribosome stops translation at that point, producing a truncated CTLA-4 protein that almost certainly loses its function. Nonsense and frameshift mutations are generally the most damaging classes.

QUIZ

CASE 4

Human and mouse CTLA-4 share ~76% amino acid identity and ~82% similarity (from the EMBOSS Needle alignment).

Why is CTLA-4 so highly conserved across ~80 million years of mammalian evolution?

CASE 5

A deletion of a single nucleotide occurs in the middle of the *CTLA-4* coding sequence.

What type of mutation results, and why is it so damaging?

CASE 6

In the lesson, multiple sequence alignment was applied to SARS-CoV-2 and related coronaviruses (bat-CoV RaTG13, pangolin-CoV, SARS-CoV-1). **What key biological insight can sequence alignment provide during a novel outbreak like COVID-19?**

Purifying selection

High conservation over long timescales is the signature of purifying (negative) selection. Mutations in CTLA-4 reduce fitness — CTLA-4 knockout mice die of massive autoimmunity within weeks of birth. Evolution has been testing random mutations for 80 million years and eliminating the ones that damage CTLA-4 function. This is also why its binding domain is a reliable drug target.

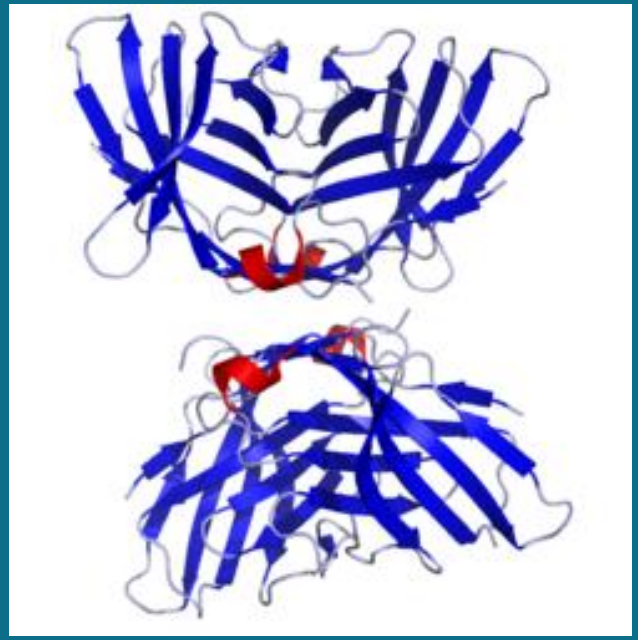
Frameshift mutation

A deletion of a single nucleotide is not divisible by 3, so it shifts the reading frame. Every codon downstream of the deletion is read incorrectly, producing an entirely wrong amino acid sequence from that point on, usually ending in a premature stop codon. This is why frameshifts are among the most damaging mutations.

Functional, Structural and Evolutionary

Sequence alignment reveals % identity between the new pathogen and known viruses, identifying the closest relatives and possible animal reservoir hosts.

The Biological Question???

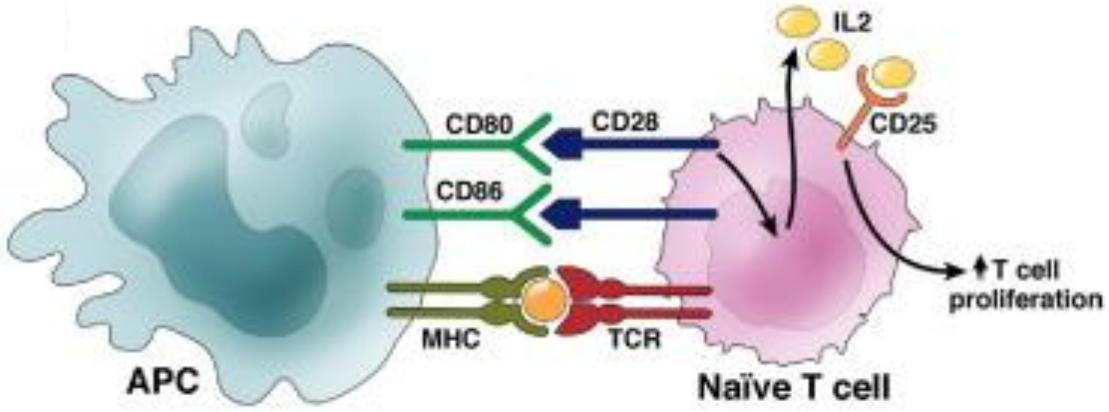


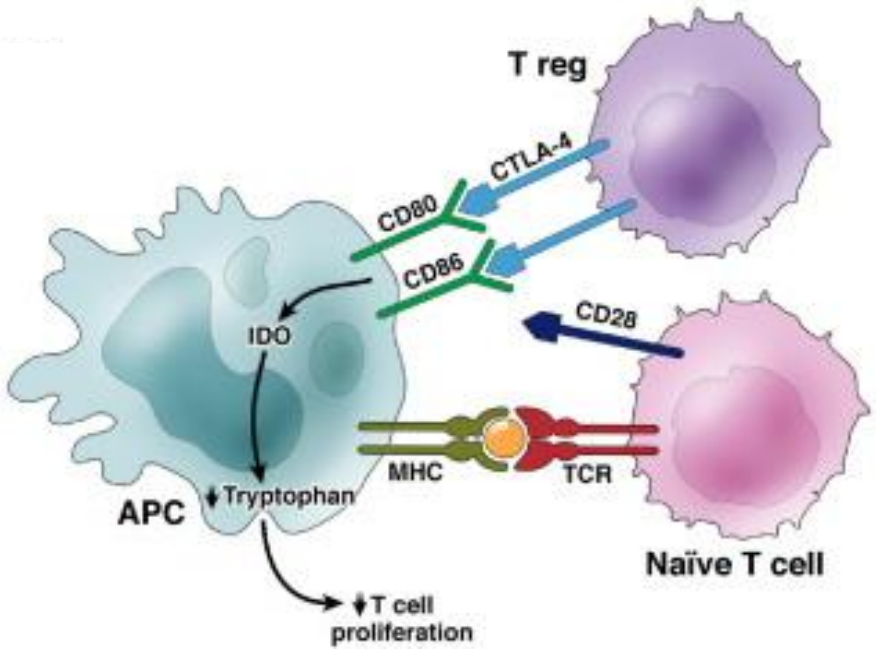
CTLA-4 (Cytotoxic T-lymphocyte-associated protein 4, also known as CD152) is a protein receptor on T cells that acts as an immune checkpoint to slow down or "turn off" the immune response.

CTLA4 (Cytotoxic T-Lymphocyte Associated Protein 4)

Activation of naive T cells requires 3 signals:

1. Presentation on MHC molecules;
2. Costimulation (CD80/CD86);
3. Stimulation by cytokines (IL-2).



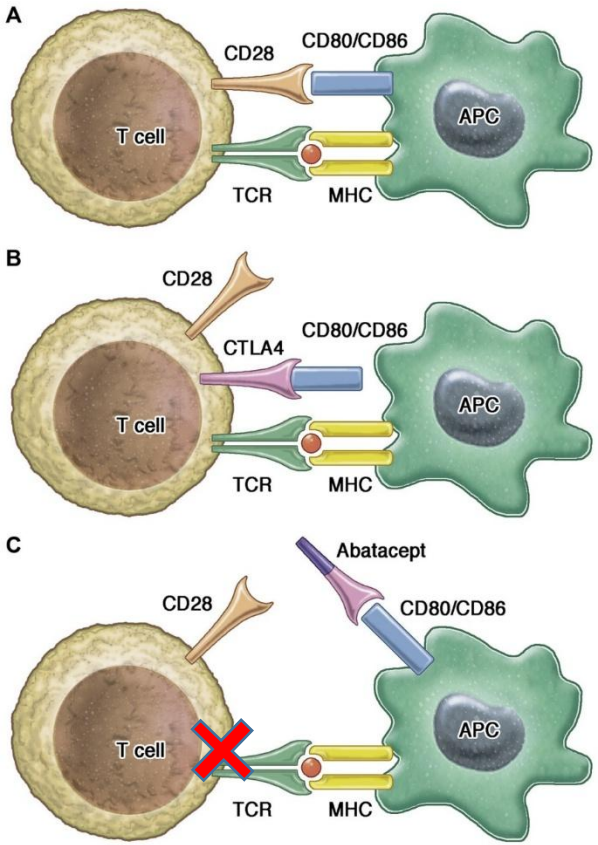


T cell proliferation

T cell proliferation

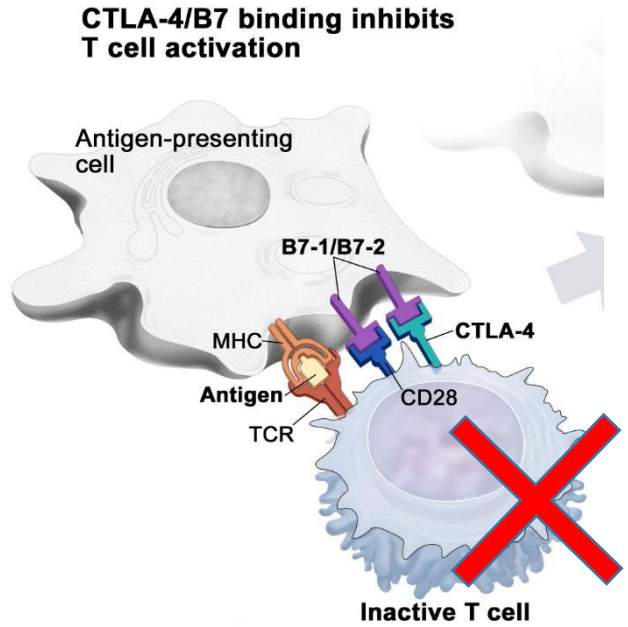
CTLA4 (Cytotoxic T-Lymphocyte Associated Protein 4)

germline loss-of-function (LoF) mutations in the *CTLA4* gene itself



Autoimmune diseases

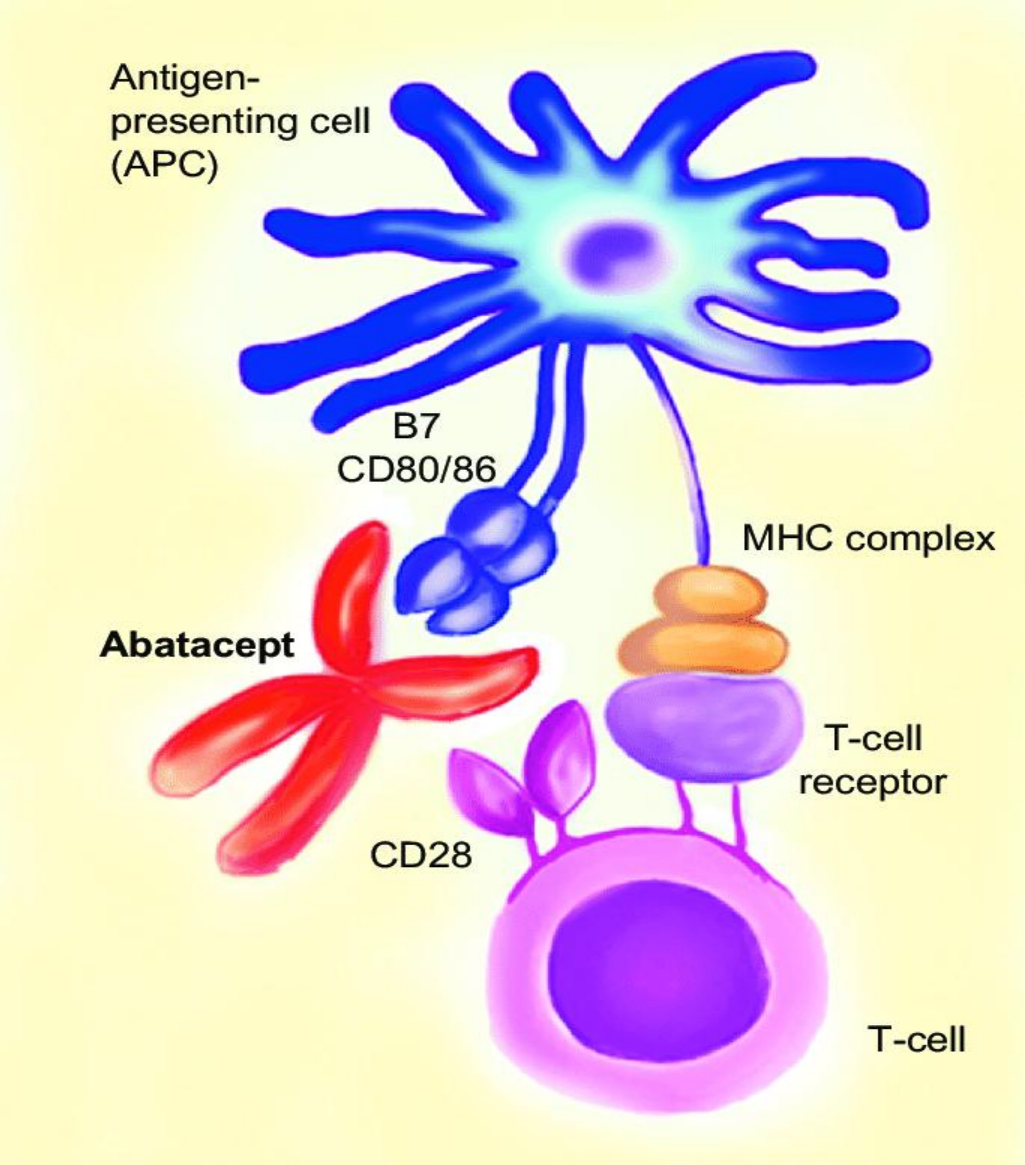
CTLA-4 expression is elevated in certain cancers

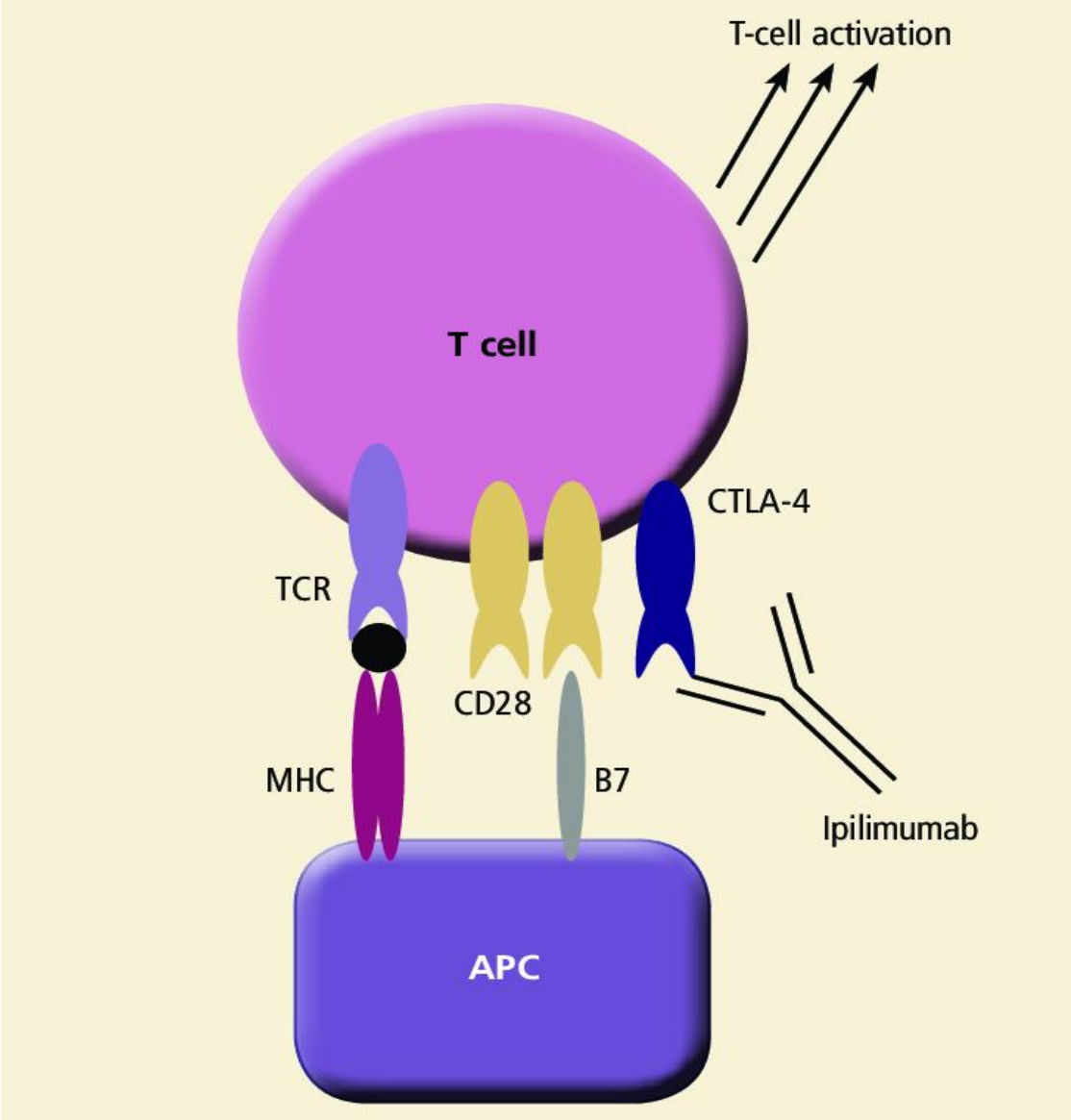




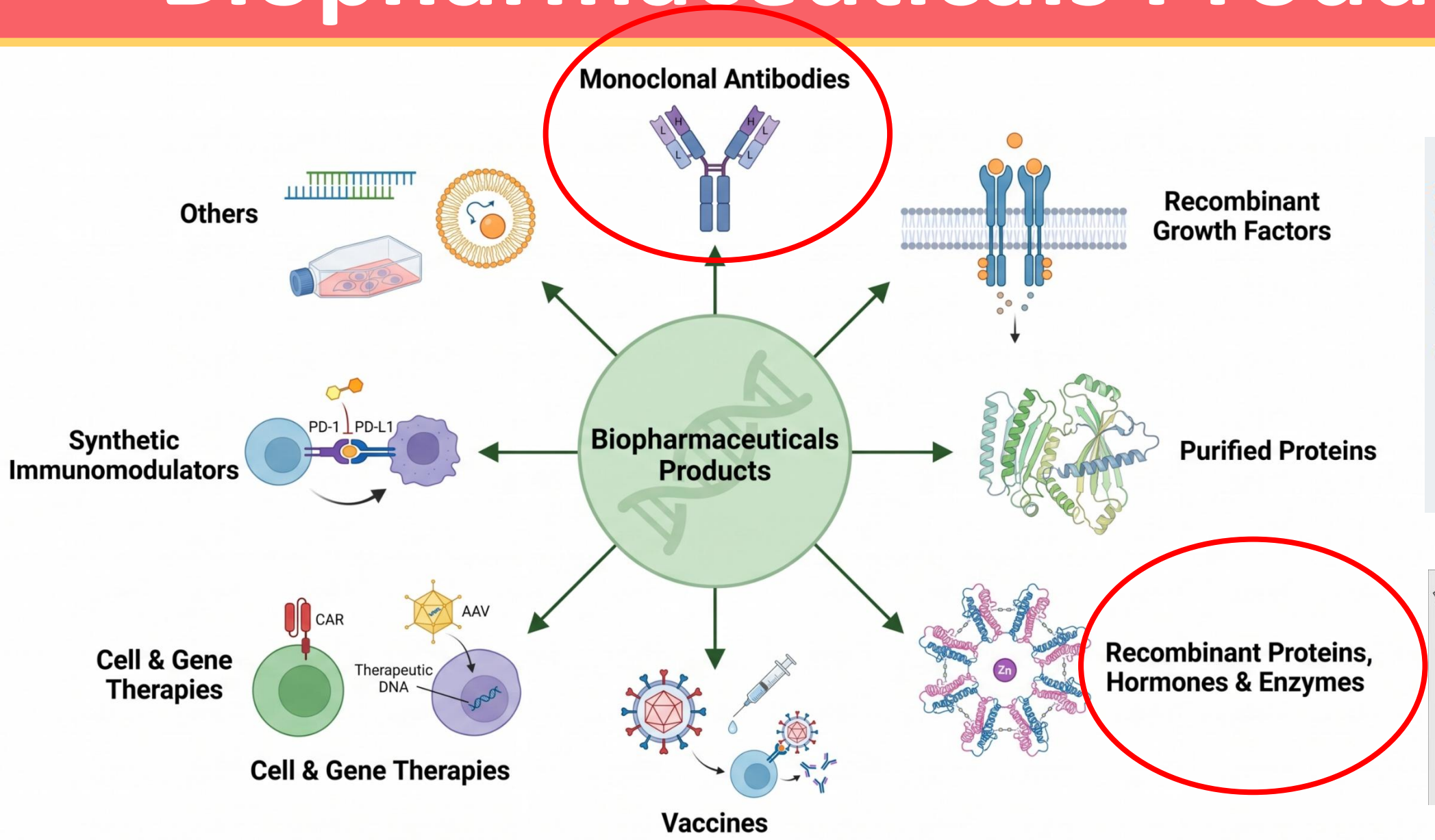
Cancer Therapy Resistance

CTLA4's Dual Role: Balancing Autoimmunity & Cancer Immunity

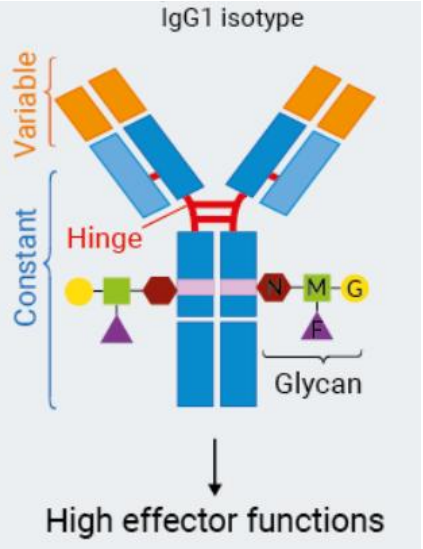


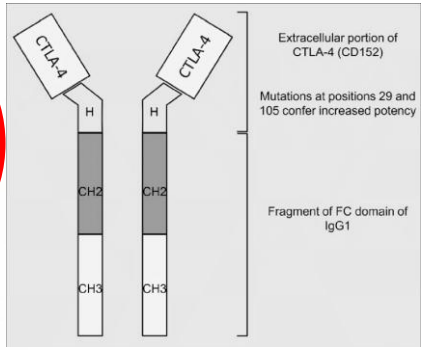


Biopharmaceuticals Products



Ipilimumab





Abatacept



“How does a biopharmaceutical reshape gene expression, and how can transcriptomics help us understand its impact?”

“Transcriptomics gives us a genome-wide snapshot of which genes are active, dysregulated, or silenced in disease”

The OMICS Landscape — Building on the Central Dogma

GenOMICS



DNA & genetic variation

Studies entire genome; SNPs, CNVs, structural variants. Foundation of precision medicine.

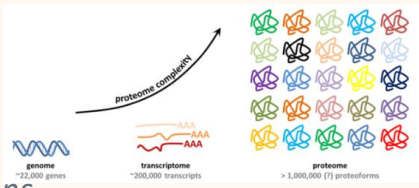
TranscriptOMICS



RNA expression

Quantifies mRNA and non-coding RNAs. Reveals gene activity in a specific cell state.

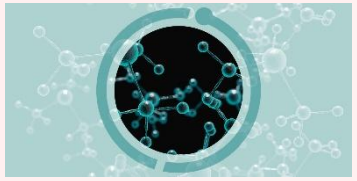
ProteOMICS



Proteins & interactions

Identifies and quantifies proteins; maps interaction networks. Closest to biological function.

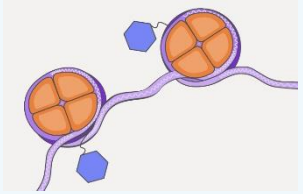
MetabolOMICS



Metabolites & pathways

Profiles small molecules. Reflects the phenotypic output of all upstream molecular events.

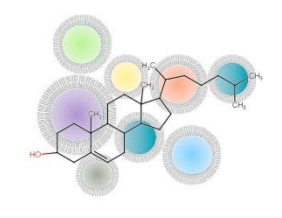
EpigenOMICS



Epigenetic modifications

Studies DNA methylation, histone modifications. Explains heritable changes without altering DNA sequence.

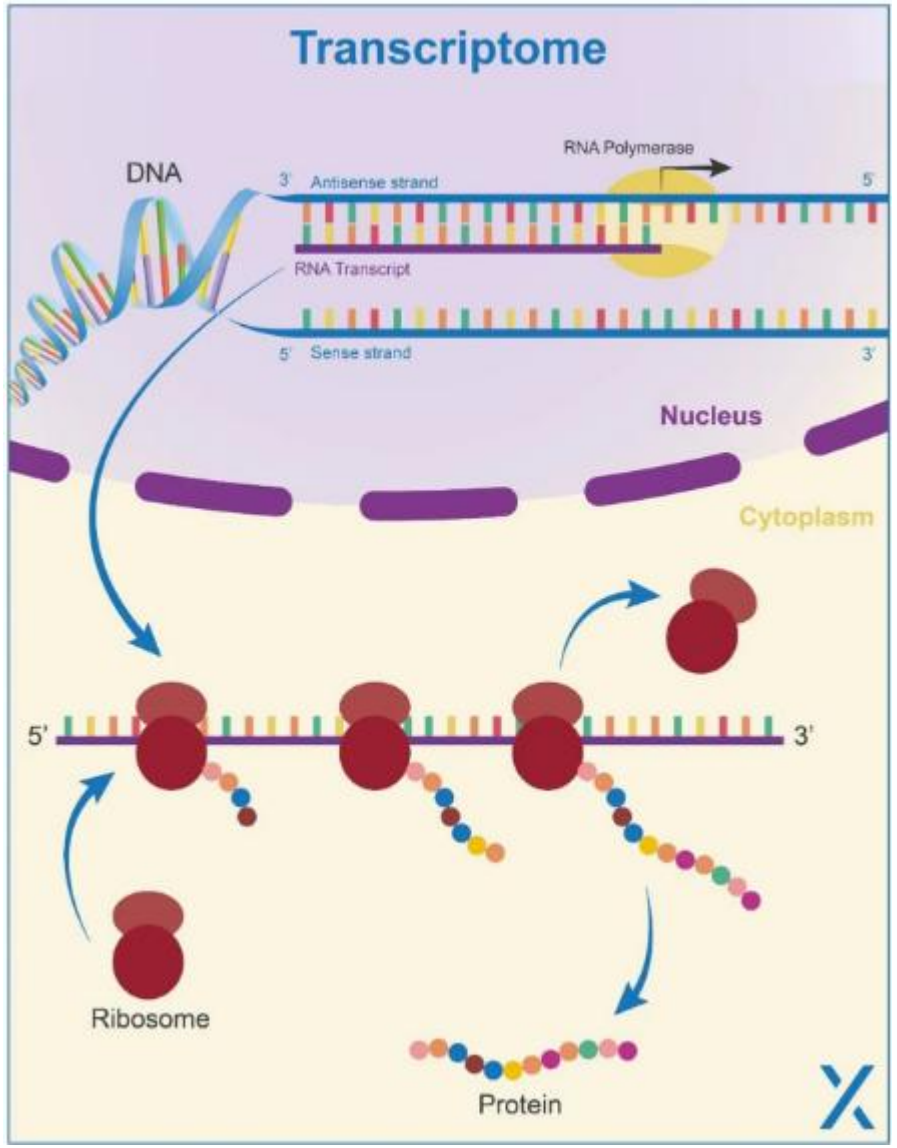
LipidOMICS



Lipid pathways & biomarkers

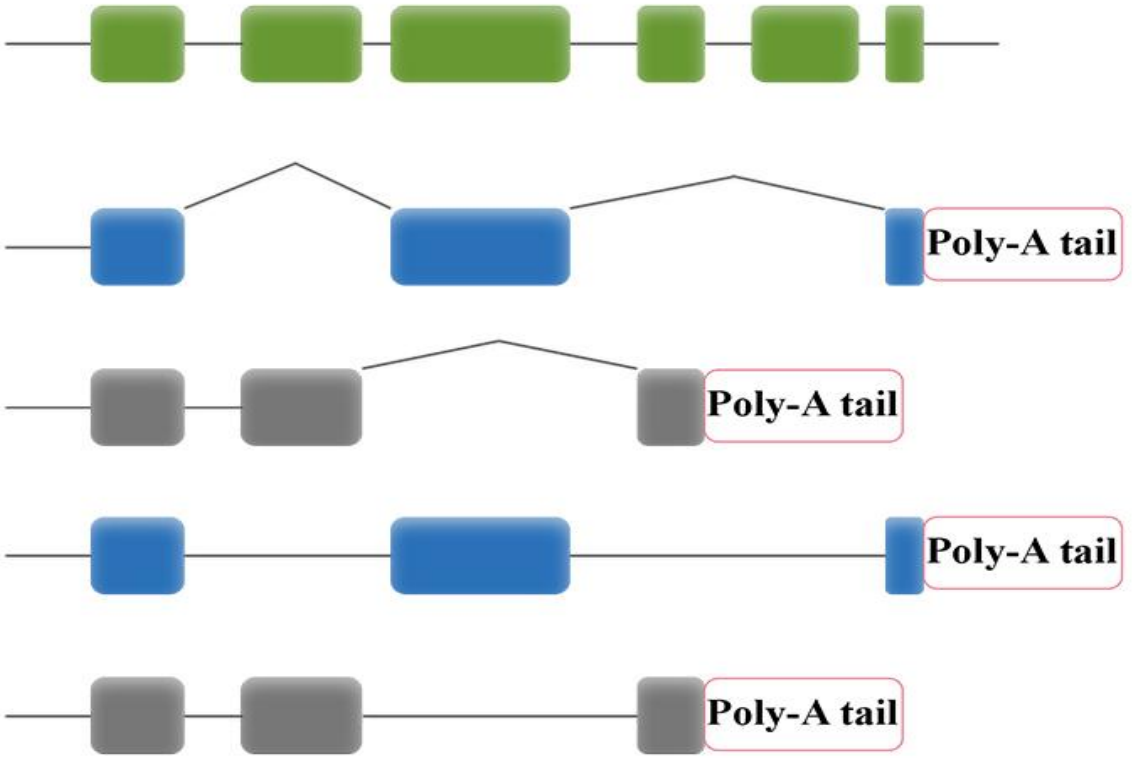
Characterises lipid species; important in cancer, cardiovascular and metabolic diseases.

Transcriptome analysis



mRNA

It directly encodes proteins



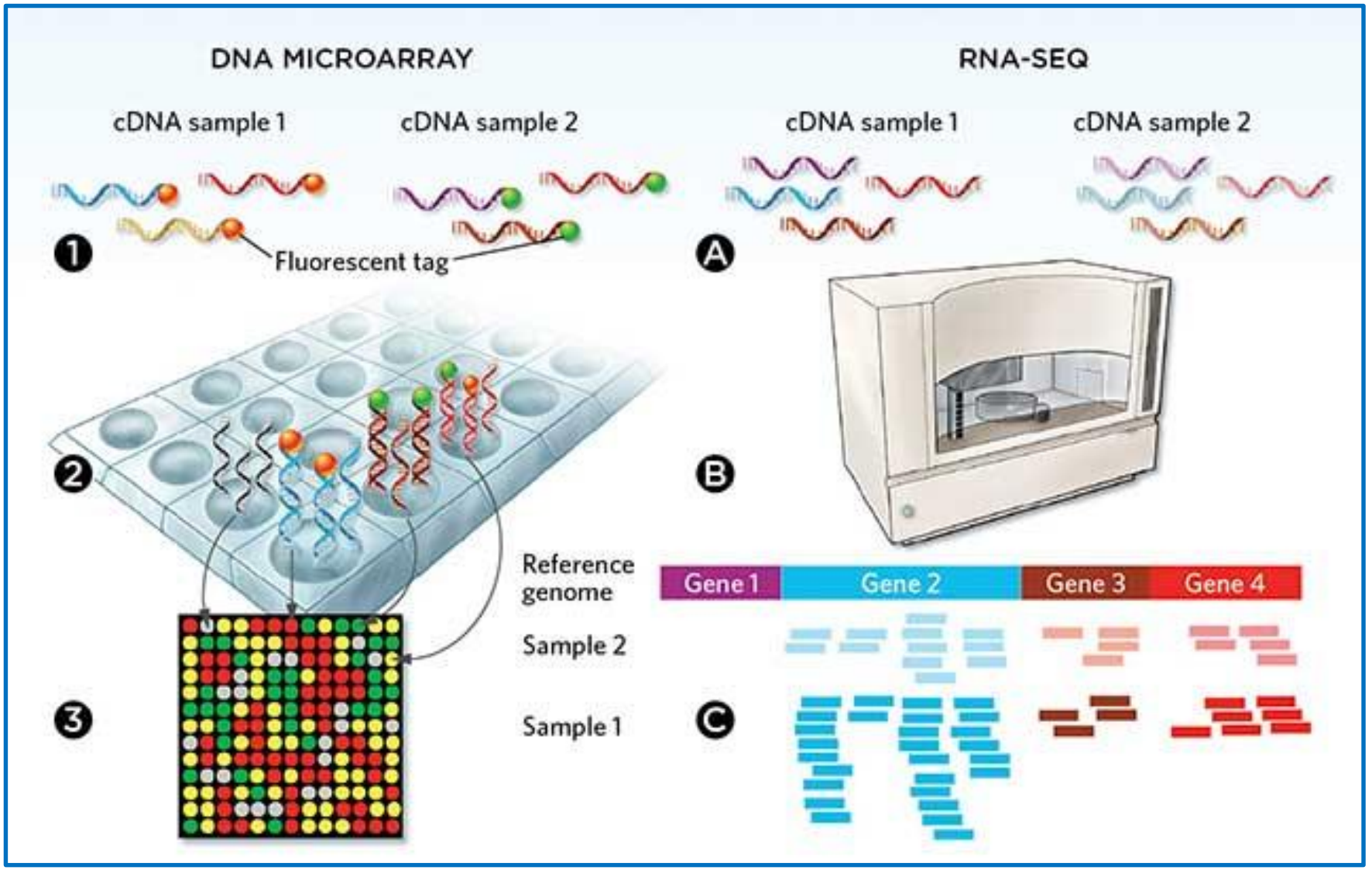
Transcriptome Techniques

Northern blotting (1970s) was the first technique developed to directly analyze RNA transcripts and study gene expression.
RT-PCR (1980s) improved sensitivity by converting RNA into cDNA and amplifying it, allowing detection of very small amounts of transcripts.



However, these techniques can analyze only a small number of transcripts at a time.

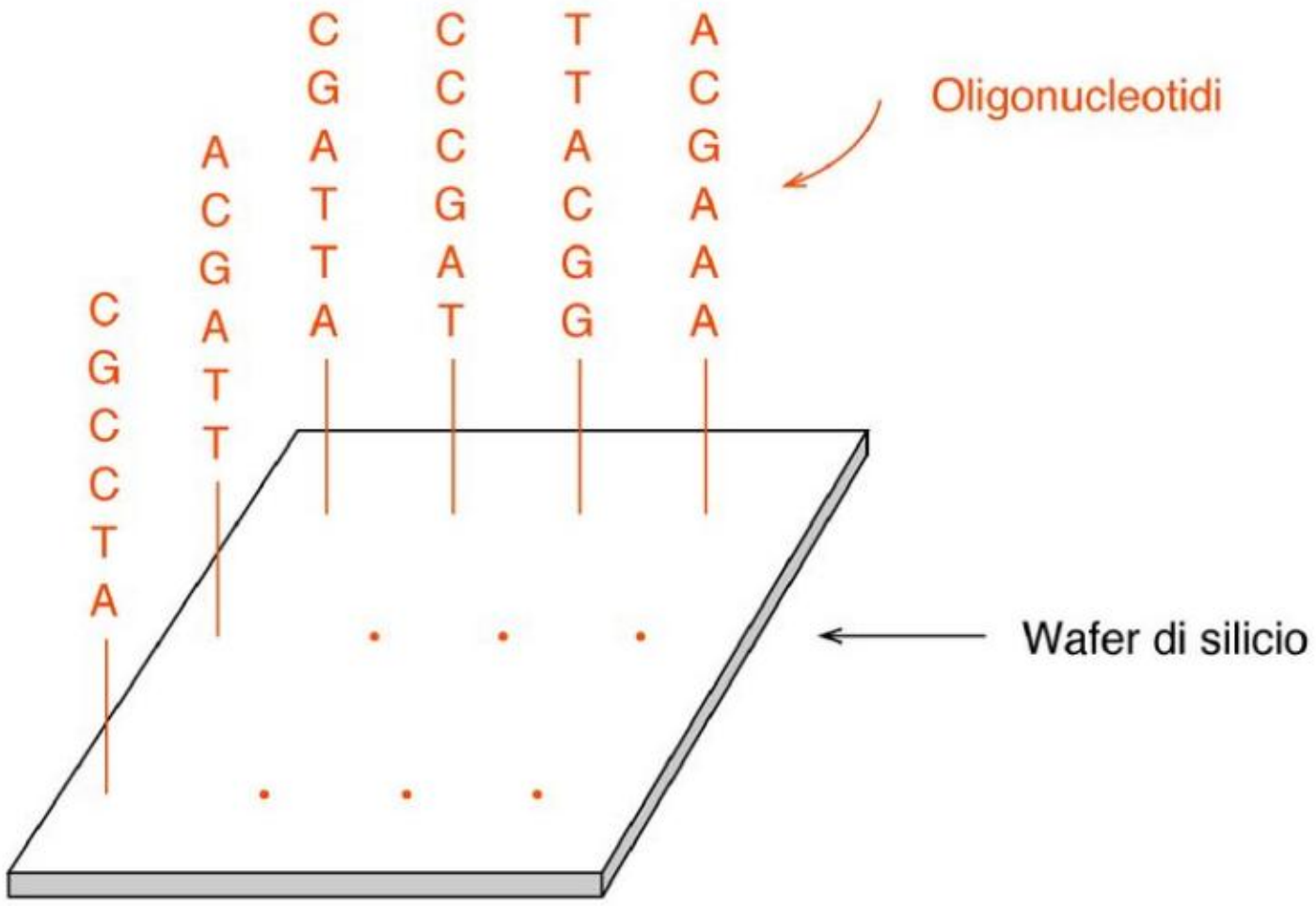
Transcriptome Techniques



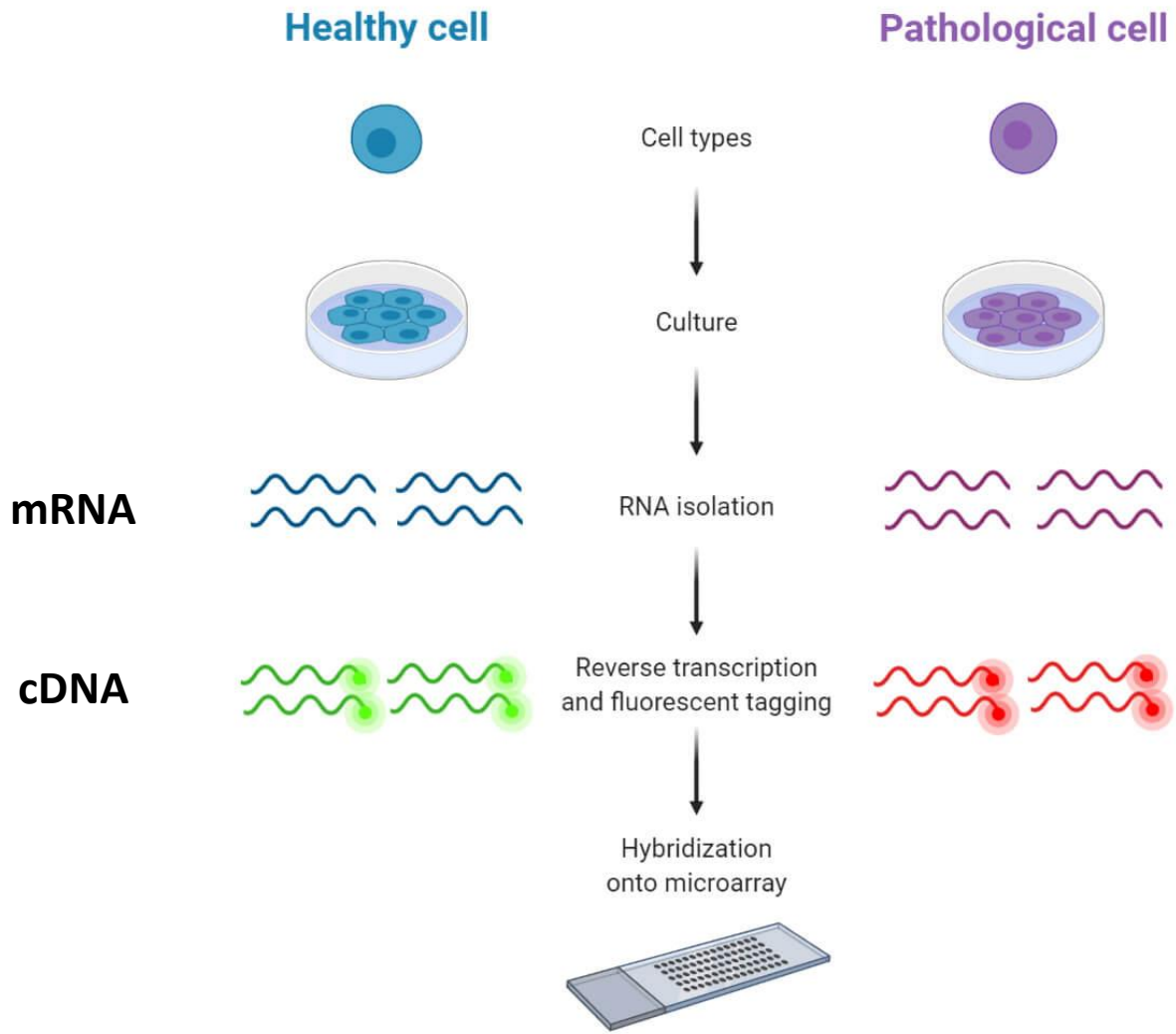
https://www.the-scientist.com/an-array-of-options-35381

Microarray

- ***Array Chip***
  - High-density microarray with surface-synthesized oligos (1 million/cm2).
  - One or more oligos correspond to an expressed DNA (gene).



Microarray



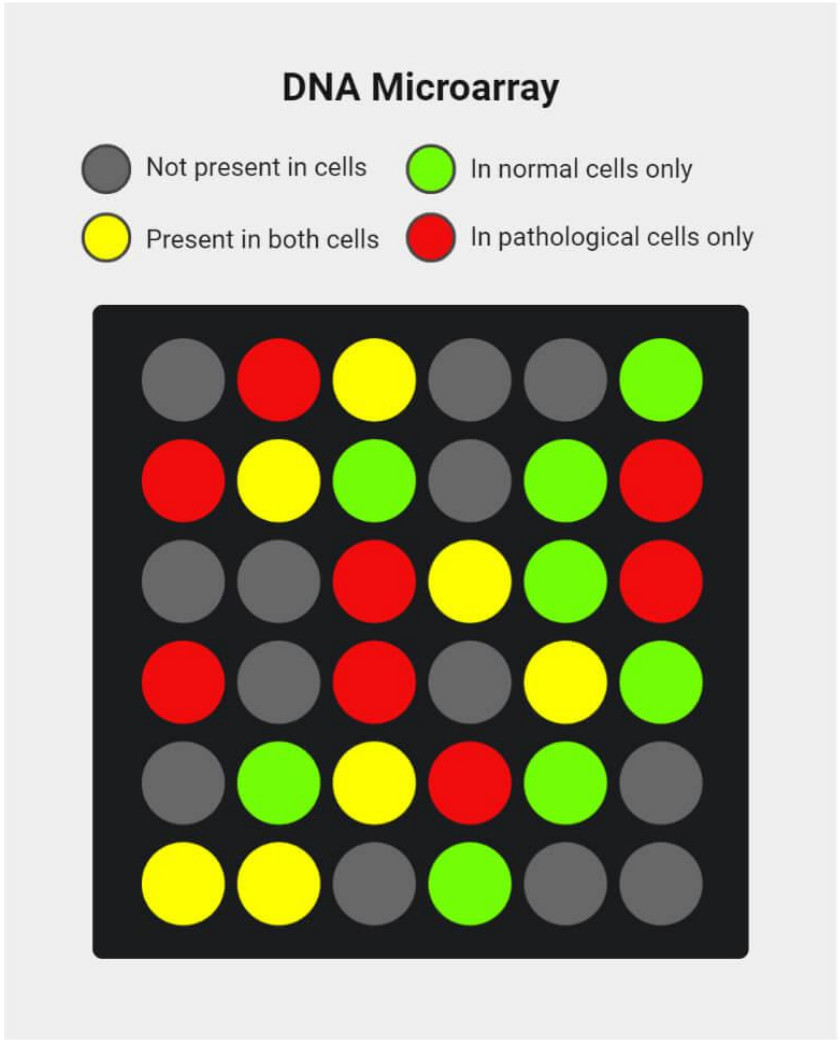


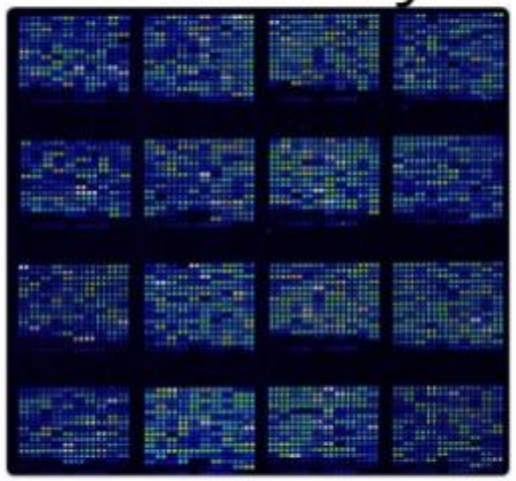
Image By Sagar Aryal, created using biorender.com

Microarray



Surescan Dx

Microarray



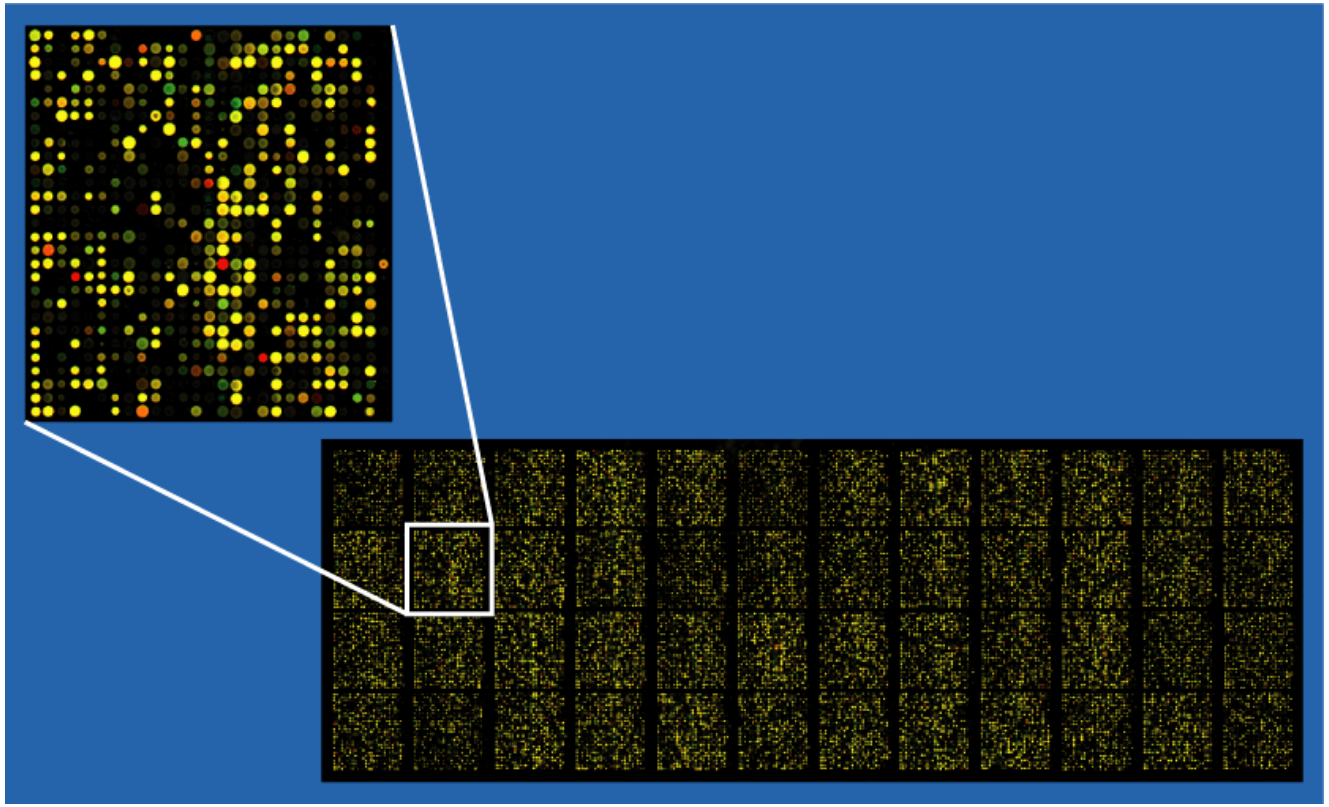


Gene chip 3000 7g

https://www.youtube.com/watch?v=wZN070rl7VA&t=139s

Microarray

The fluorescence intensity in the region of each probe indicates the abundance of the respective transcript.



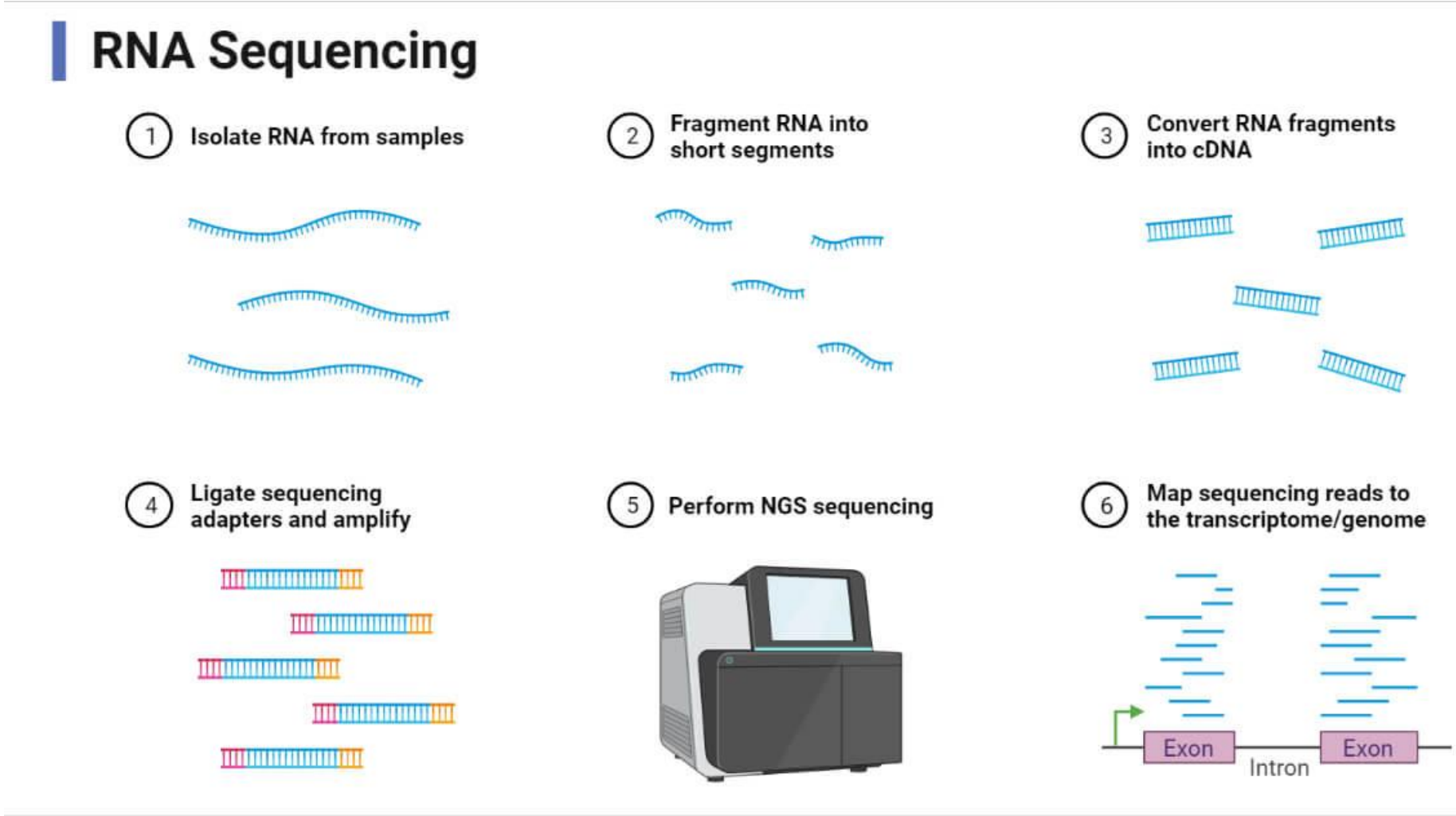
Limitations:

- High noise levels,
- Possibility of saturation,
- Depends on the existence of a reference genome for probe selection/design.

Analysis of thousands of transcripts simultaneously and increased sensitivity and accuracy in measuring low-abundance transcripts.

RNA sequencing

High-throughput sequencing with single-base resolution combined with computational methods for transcript quantification.

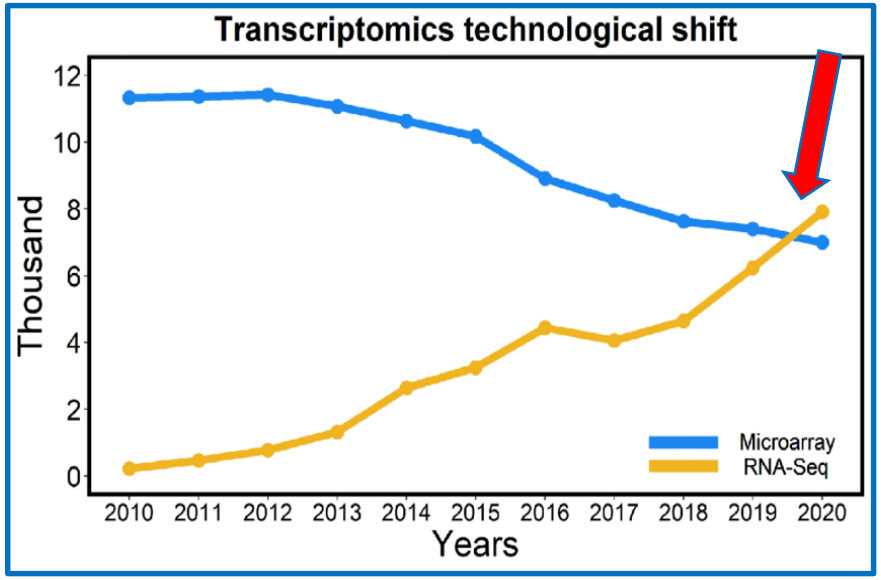


RNA sequencing

Using Next Generation Sequencing (NGS) technology, RNA sequencing (RNA-seq) continues to expand the frontiers of biology.

Technology	Advantages	Limitations
qPCR Array	Low-cost; simple	Only testing limited number of genes of interest in specific pathways
Microarray	Low-cost; ability to process large number of samples; high-throughput	Low sensitivity for very lowly-or very highly expressed genes; high background; difficult to detect novel transcripts
RNA-seq	High accuracy; high sensitivity and dynamic range; low background/noise signal; high-throughput; identify novel transcripts, splice junctions, SNPs and non-coding RNAs	High-cost; high data storage

SNPs, single nucleotide polymorphism.



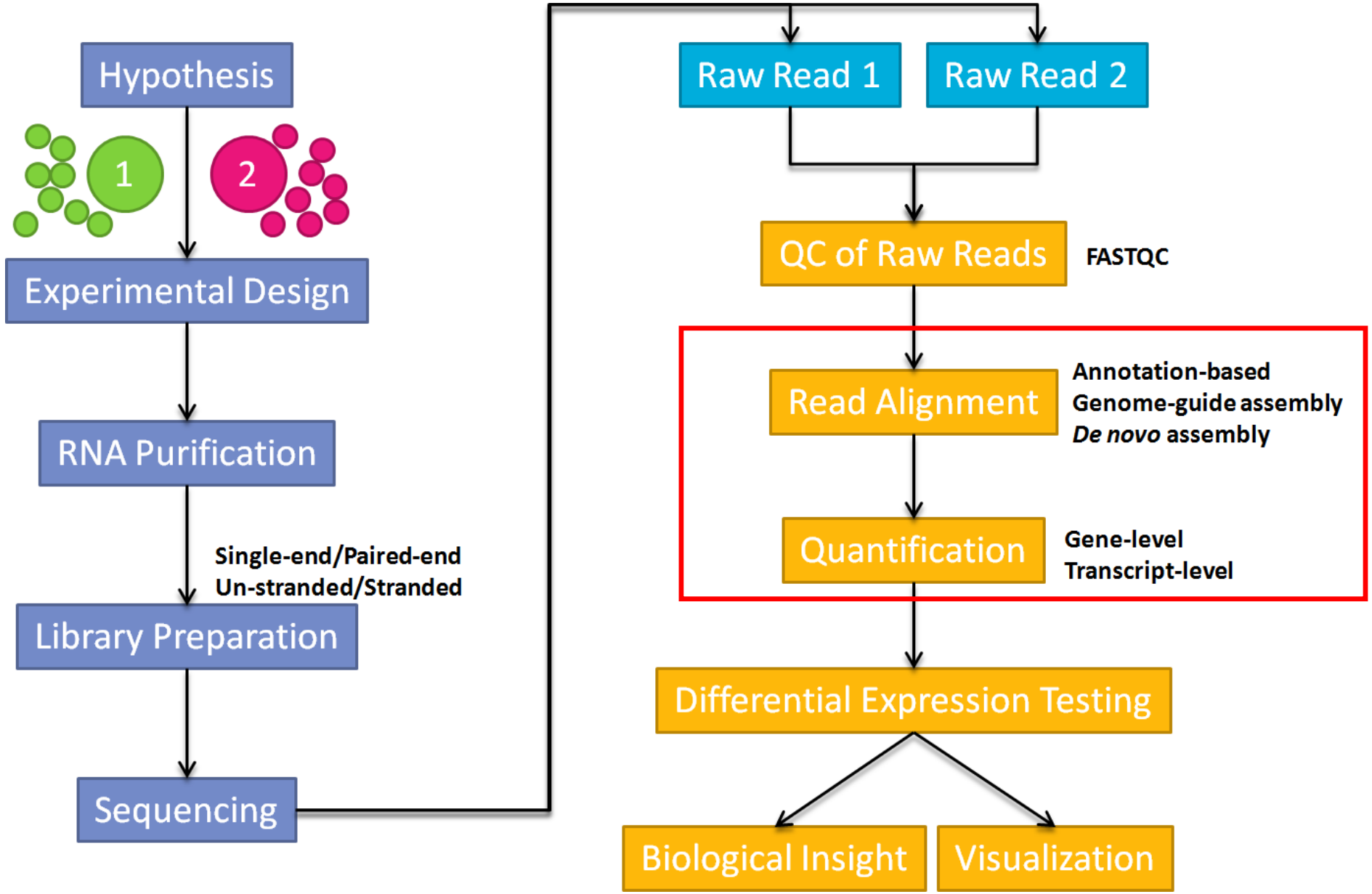
Xu S., Front Pharmacol. 2017

Challenges:

- Sample handling stages
- Difficulty in processing and analyzing transcripts
- Higher costs?

Applications:

- Gene expression
- Splicing diversity
- Pathogen virulence factors
- Antibiotic resistance


Hypothesis
1
2
Experimental Design
RNA Purification
Single-end/Paired-end
Un-stranded/Stranded
Library Preparation
Sequencing
Raw Read 1
Raw Read 2
QC of Raw Reads
FASTQC
Read Alignment
Annotation-based
Genome-guide assembly
De novo assembly
Quantification
Gene-level
Transcript-level
Differential Expression Testing
Biological Insight
Visualization

RNA sequencing

Sequencing count data

	control-1	control-2	control-3	treated-1	treated-2
FBgn0000008	78	46	43	47	89
FBgn0000014	2	0	0	0	0
FBgn0000015	1	0	1	0	1
FBgn0000017	3187	1672	1859	2445	4615
FBgn0000018	369	150	176	288	383
[...]					

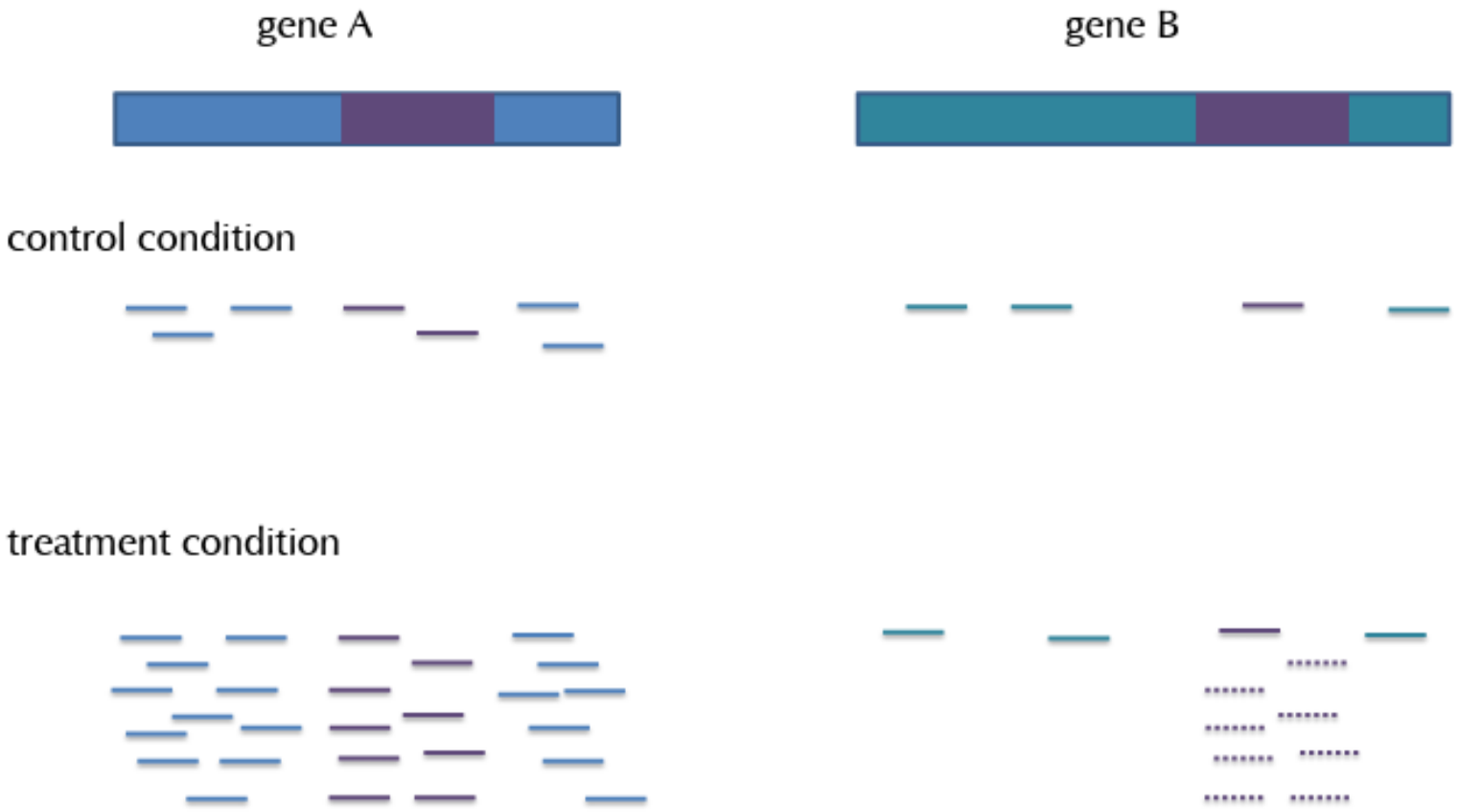
- RNA-Seq
- Tag-Seq
- ChIP-Seq
- HiC
- Bar-Seq
- ...

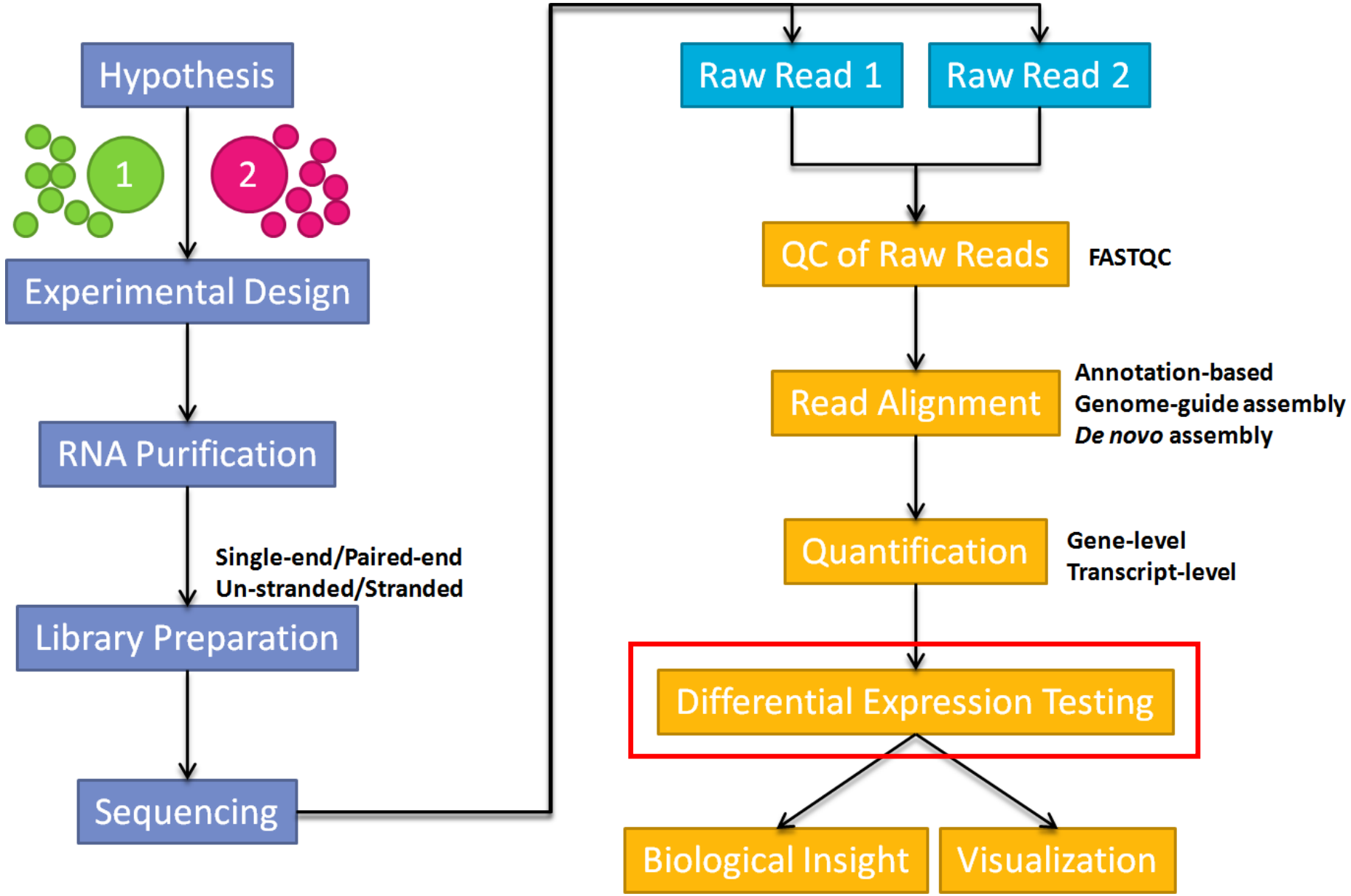
Counting rules

- Count reads, not base-pairs
- Count each read at most once.
- Discard a read if
  - it cannot be uniquely mapped
  - its alignment overlaps with several genes
  - the alignment quality score is bad
  - (for paired-end reads) the mates do not map to the same gene

RNA sequencing

Why we discard non-unique alignments




Hypothesis
1
2
Experimental Design
RNA Purification
Single-end/Paired-end
Un-stranded/Stranded
Library Preparation
Sequencing
Raw Read 1
Raw Read 2
QC of Raw Reads
FASTQC
Read Alignment
Annotation-based
Genome-guide assembly
De novo assembly
Quantification
Gene-level
Transcript-level
Differential Expression Testing
Biological Insight
Visualization

RNA sequencing - normalization



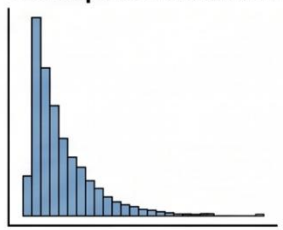
DESeq2 Workflow: Negative Binomial GLM for RNA-seq Differential Expression Analysis

The Problem: Count Data from RNA-seq

Raw Count Matrix

	Ctrl1	Ctrl2	Ctrl3	Trt1	Trt2	Trt3
Gene A	0	5	234	1500	87	92
Gene B	0	5	234	500	87	92
Gene C	0	5	198	290	67	48
...	...	...	...	...	...	...
Gene N	0	1	46	33	10	14

Overdispersed Distribution

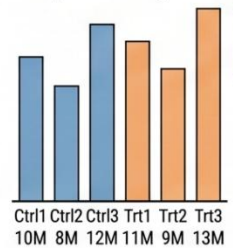


Example counts: 0, 5, 234, 1500

Raw counts are overdispersed integers — cannot use t-test or ANOVA

Step 1: Normalization by Size Factors

Unequal Library Sizes

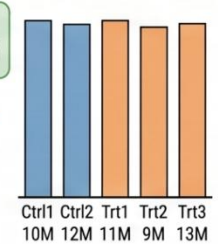


Total Reads

$$s_j = \frac{\text{median}(K_{ij})}{\text{geometric_mean}(K_i)}$$

Corrects for technical variation in sequencing depth and composition

Normalized Counts

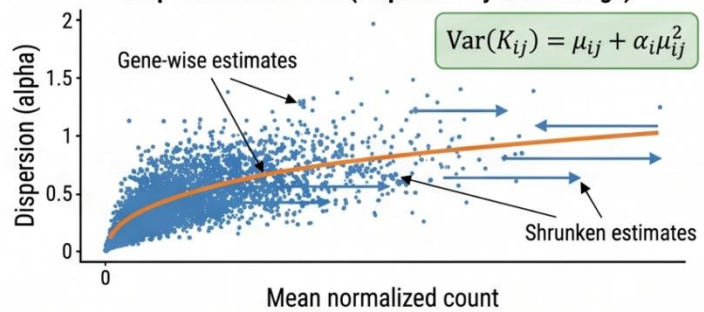


Adjusted Total Reads

Median of ratios normalization — accounts for library size AND composition bias

Step 2: Dispersion Estimation

Dispersion-Mean Plot (Empirical Bayes Shrinkage)



Negative Binomial model — captures overdispersion in count data. Empirical Bayes shrinkage pools information across genes

Step 3: Negative Binomial GLM Fitting

$$K_{ij} \sim \text{NB}(\mu_{ij}, \alpha_i)$$
$$\log(\mu_{ij}) = \sum x_{jr}\beta_{ir}$$

Design Matrix (X)

	Intercept	Condition
Ctrl1	1	0
Ctrl2	1	0
Ctrl3	1	0
Trt1	1	1
Trt2	1	1
Trt3	1	1

Coefficient Estimates

$$\begin{bmatrix} \text{Intercept } \beta_0 \\ \text{Condition } \beta_1 \end{bmatrix} \rightarrow \text{log2 fold change}$$

Generalized Linear Model with log link function — estimates log2 fold change for each gene

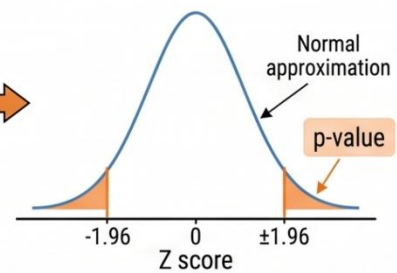
Step 4: Wald Test for Significance

Wald Test Hypotheses

H0: β = 0
H1: β ≠ 0

$$W = \frac{\hat{\beta}}{\text{SE}(\hat{\beta})}$$

Wald Statistic Distribution Under H0



Wald test: tests if log2FC is significantly different from zero. Alternative: LRT (Likelihood Ratio Test) for complex designs

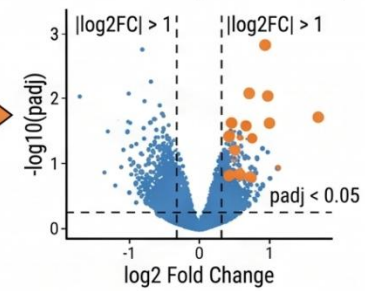
Step 5: Multiple Testing Correction

20,000 genes tested → ~1,000 expected false positives at p<0.05 by chance

p-values
Rank p-values
Compute Adjusted p-values
padj

$$padj = p \times \frac{m}{rank}$$

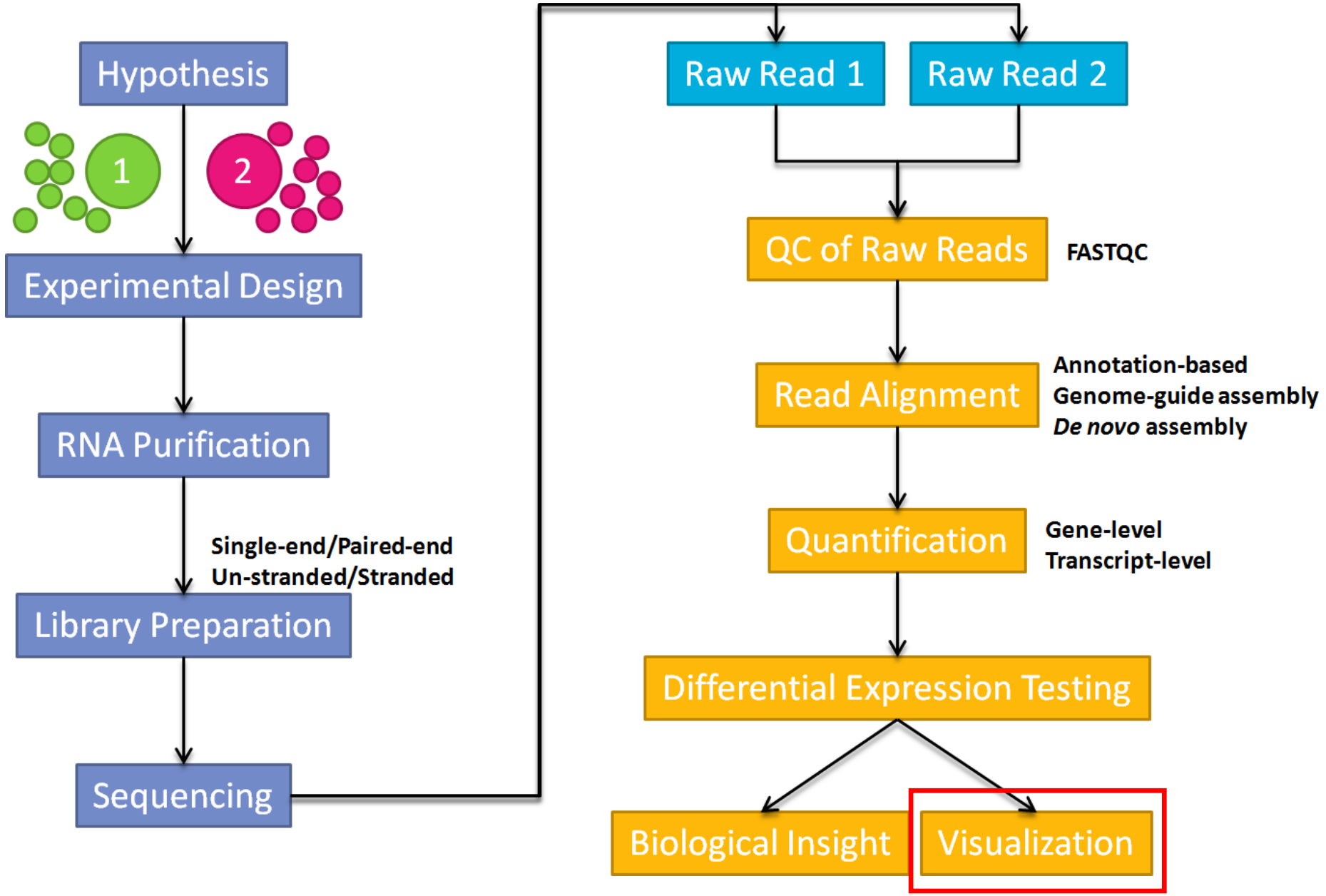
Volcano Plot (BH Correction)



BH correction controls False Discovery Rate (FDR). Adjusted p-value (padj) is the key output

Key assumptions

1. Counts follow Negative Binomial distribution.
2. Size factors capture systematic technical variation.
3. Dispersion is shared across similar genes.
4. Sufficient replicates per condition (minimum n=3).


Hypothesis
1
2
Experimental Design
RNA Purification
Single-end/Paired-end
Un-stranded/Stranded
Library Preparation
Sequencing
Raw Read 1
Raw Read 2
QC of Raw Reads
FASTQC
Read Alignment
Annotation-based
Genome-guide assembly
De novo assembly
Quantification
Gene-level
Transcript-level
Differential Expression Testing
Biological Insight
Visualization

Statistical Inference

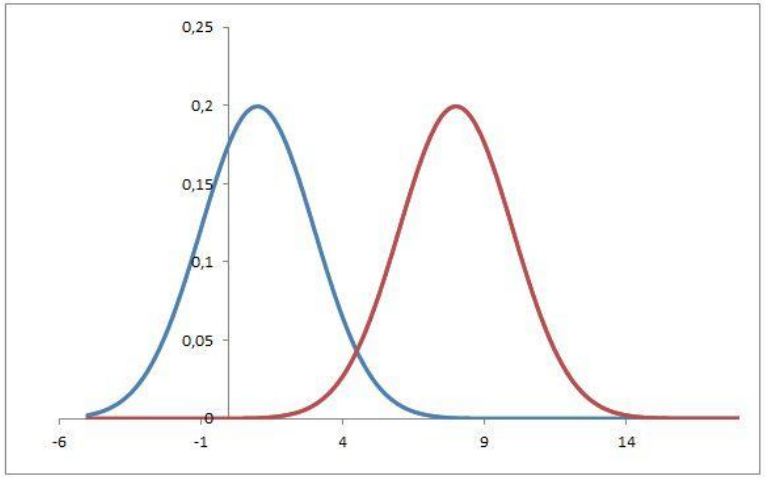
- It's often necessary to confirm the values of a variable in two populations. In this case, assuming the variable is a continuous numerical variable, the question is: ***does the variable have a significantly different mean in the two populations?***
- **Hypothesis Testing**
  - The Null Hypothesis (H0)
  - The Alternative Hypothesis (H1) -> Denial of the Null Hypothesis

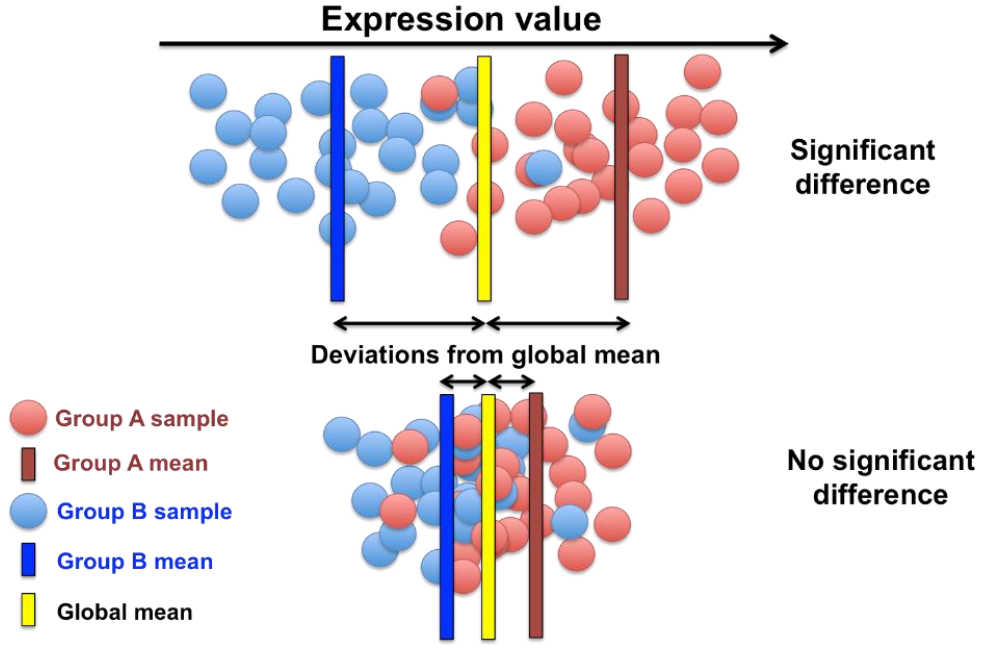
$$H_0 : \mu_1 = \mu_2$$

$$H_0 : \mu_1 \leq \mu_2$$

Level of significance (p-value)

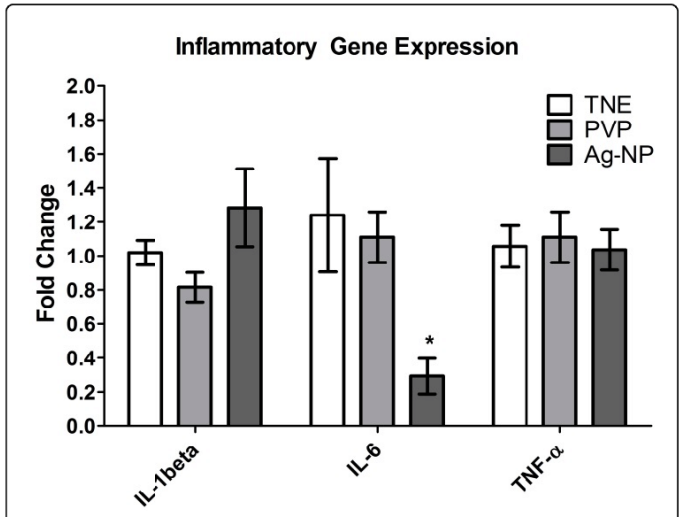
- To gain more information about the actual probability of observing a certain test value in the distribution under H0, the p-value is usually reported.
- The p-value, therefore, is a probability with values ranging from 0 to 1: small p-values lead to the rejection of H0.





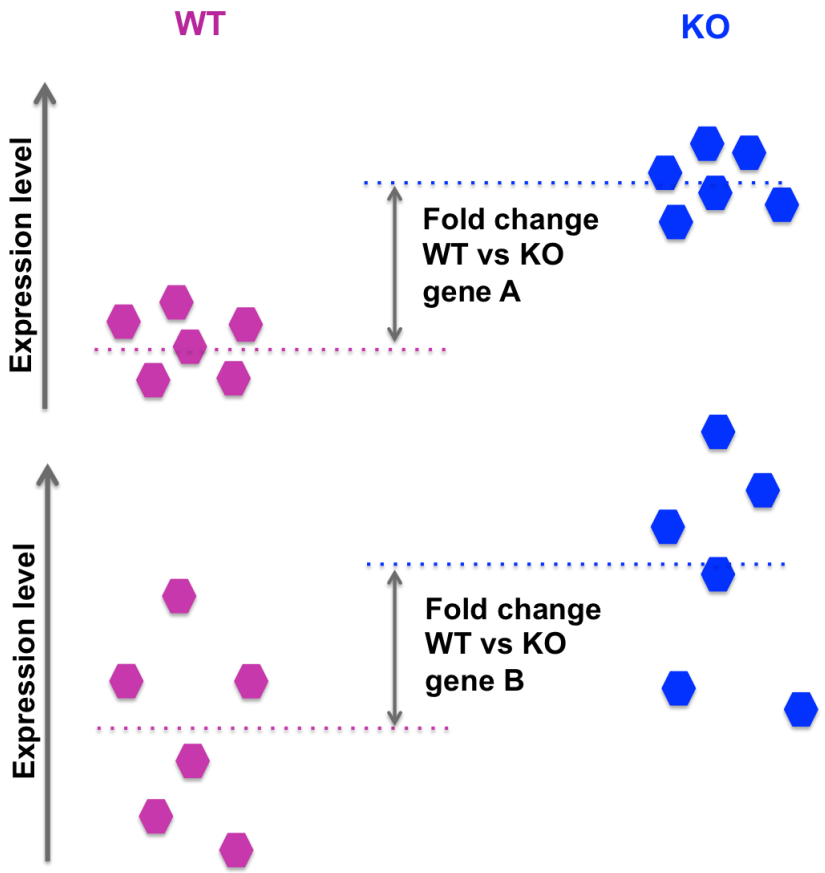
Fold Change

- Fold change is a simple way to describe how much a value changes relative to a reference.
  - If the value doubles relative to the reference, the fold change is 2.
  - If the value halves, the fold change is 0.5.
  - A fold change of 1 means there has been no change.



Log-Fold Change (*logFC*)

- Fold change is often converted to a logarithmic scale, such as log2 fold change:
  - Log2(Fold Change) = 1 → the gene has doubled in expression.
  - Log2(Fold Change) = -1 → expression has halved.
  - Log2(Fold Change) = 0 → no change in expression.

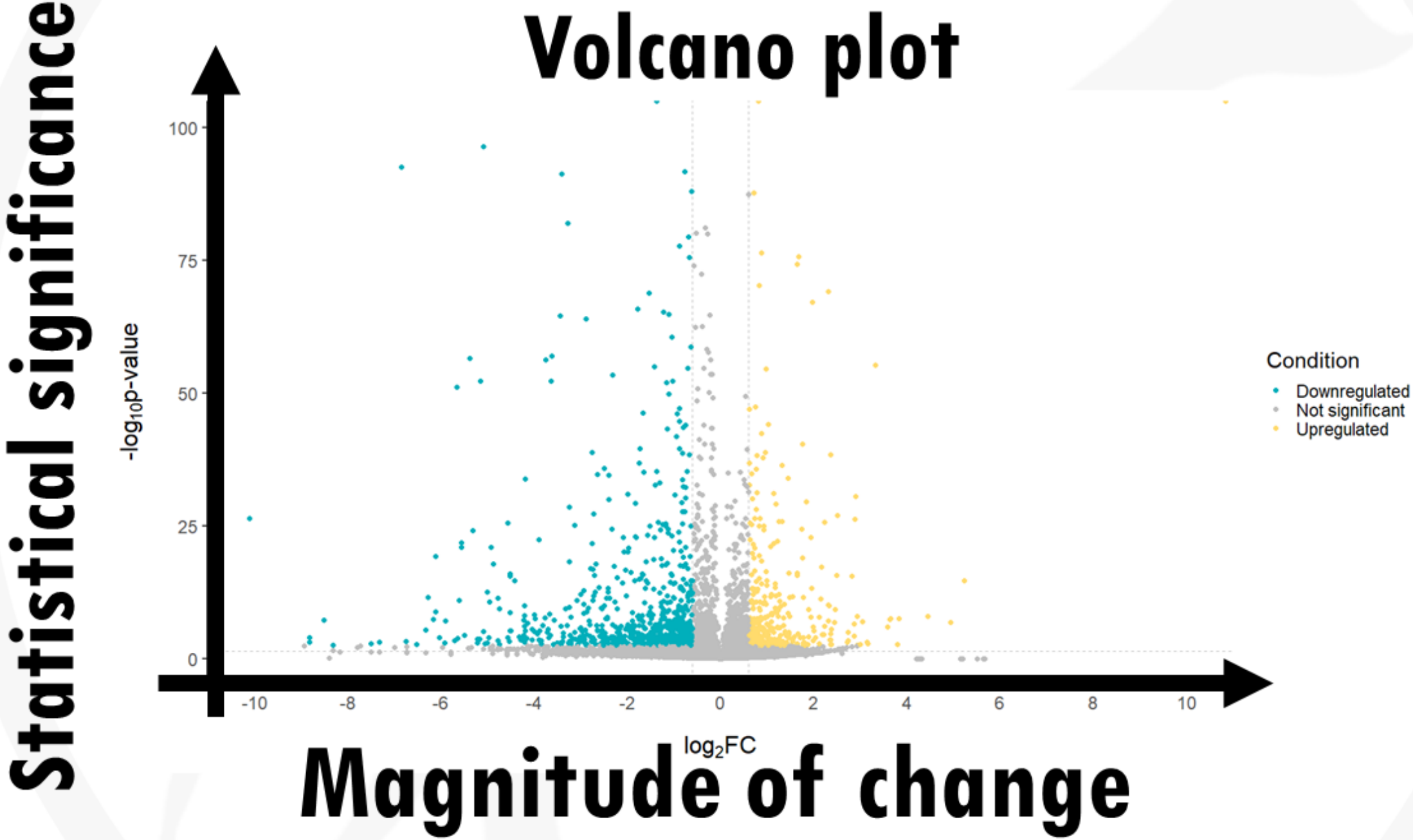


Log Base 2

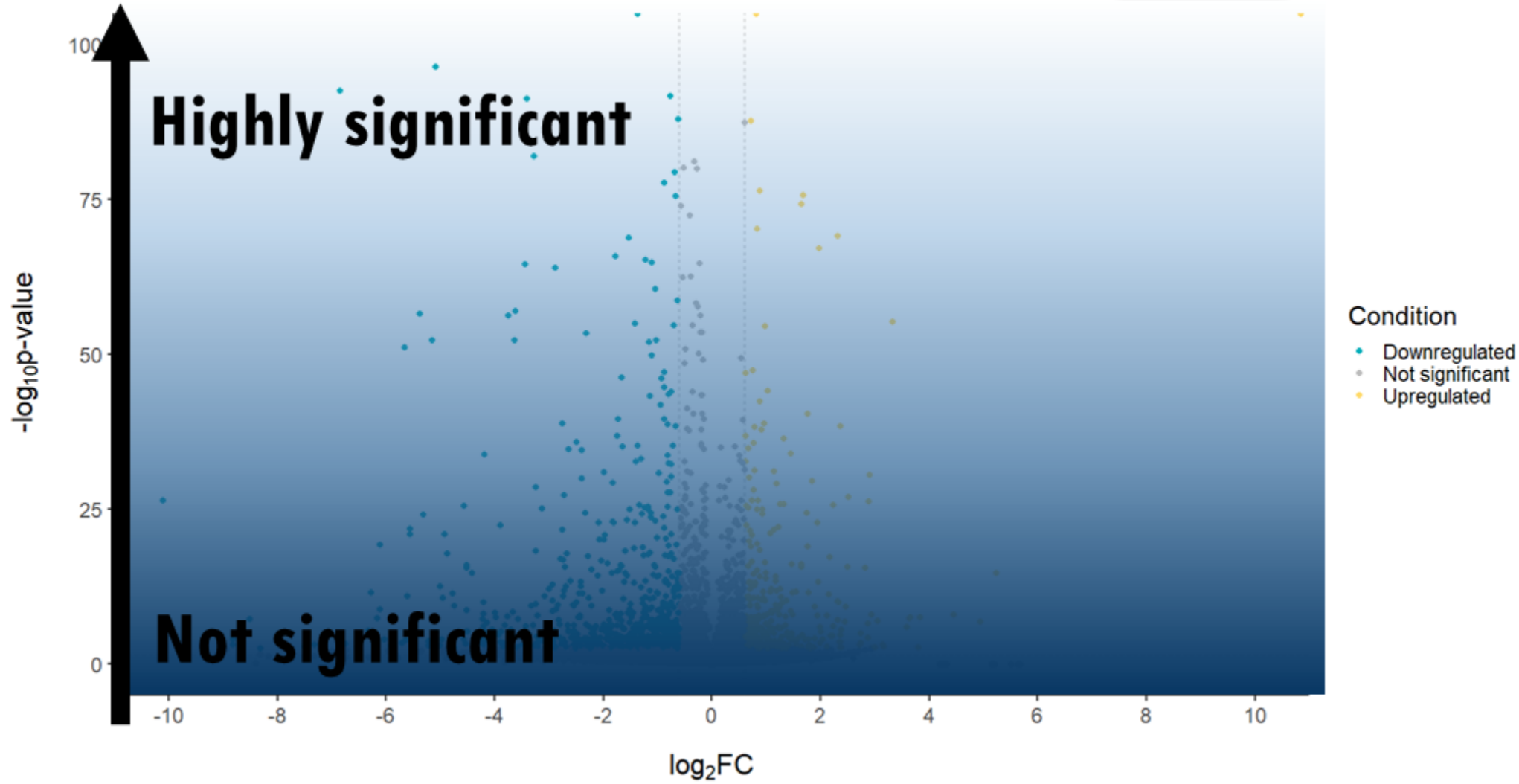
cuemath
THE MATH EXPERT

Logarithmic Form	Exponential Form
$\log_2^N = K$	$2^K = N$

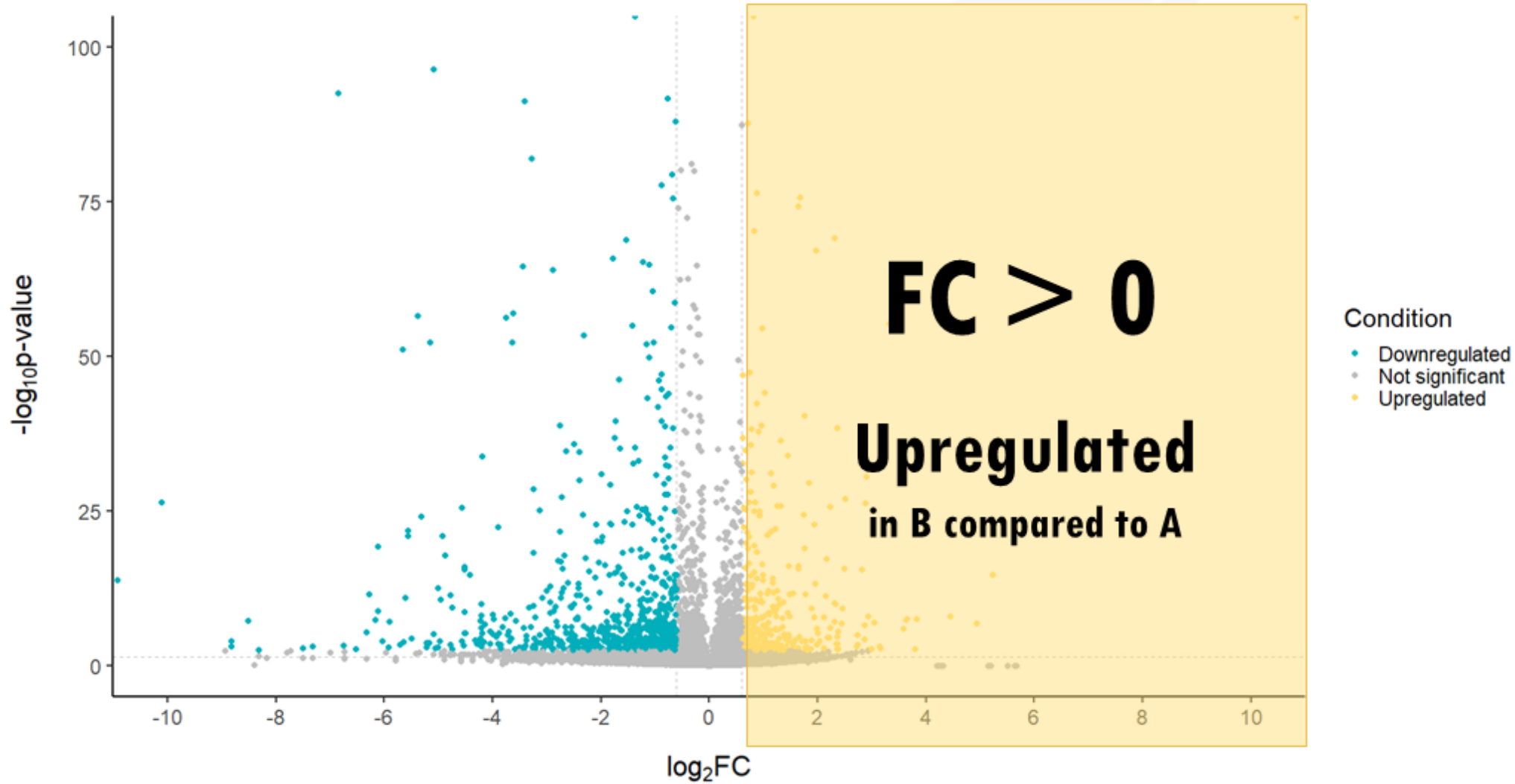
Volcano plot



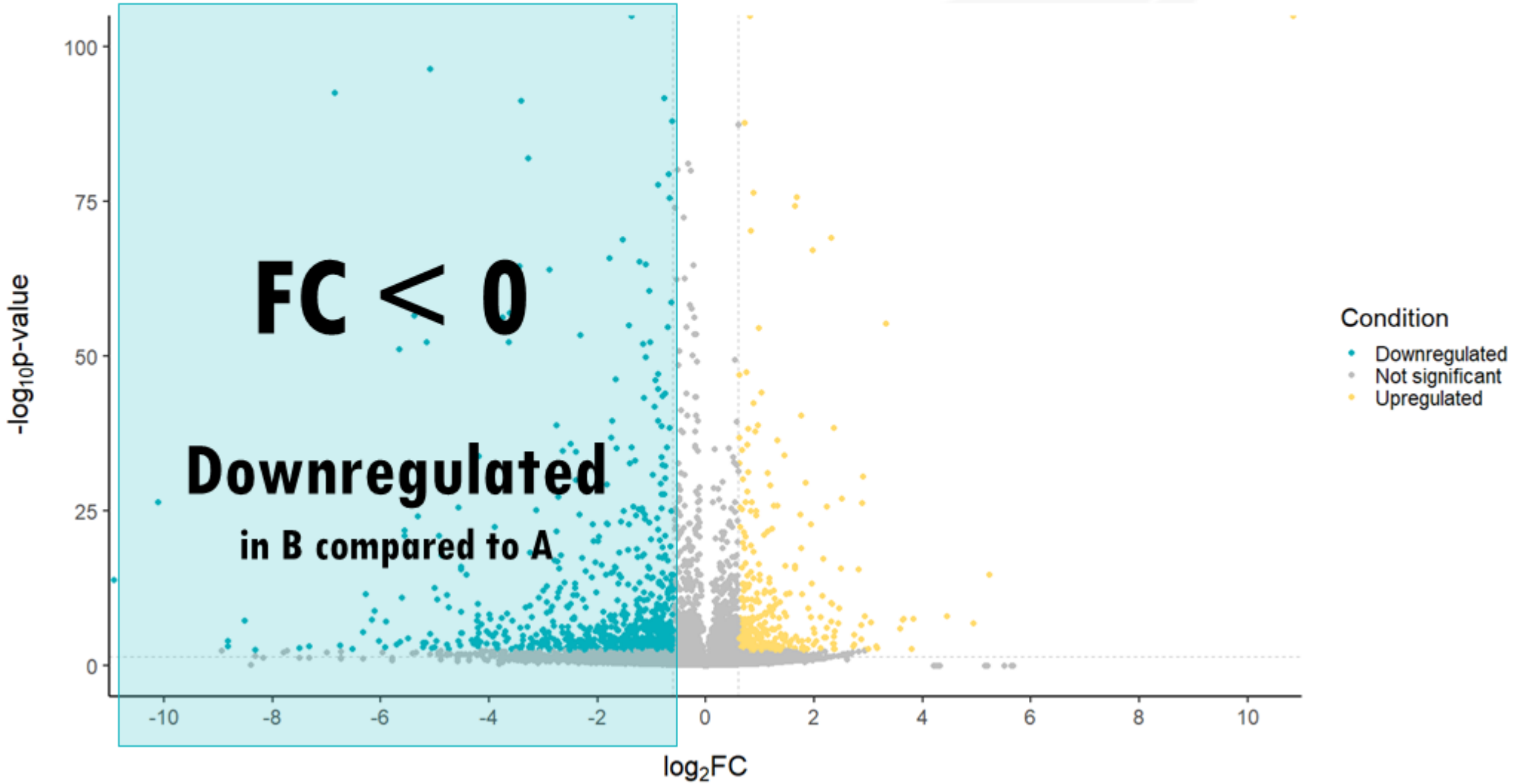
Statistical significance: p-value



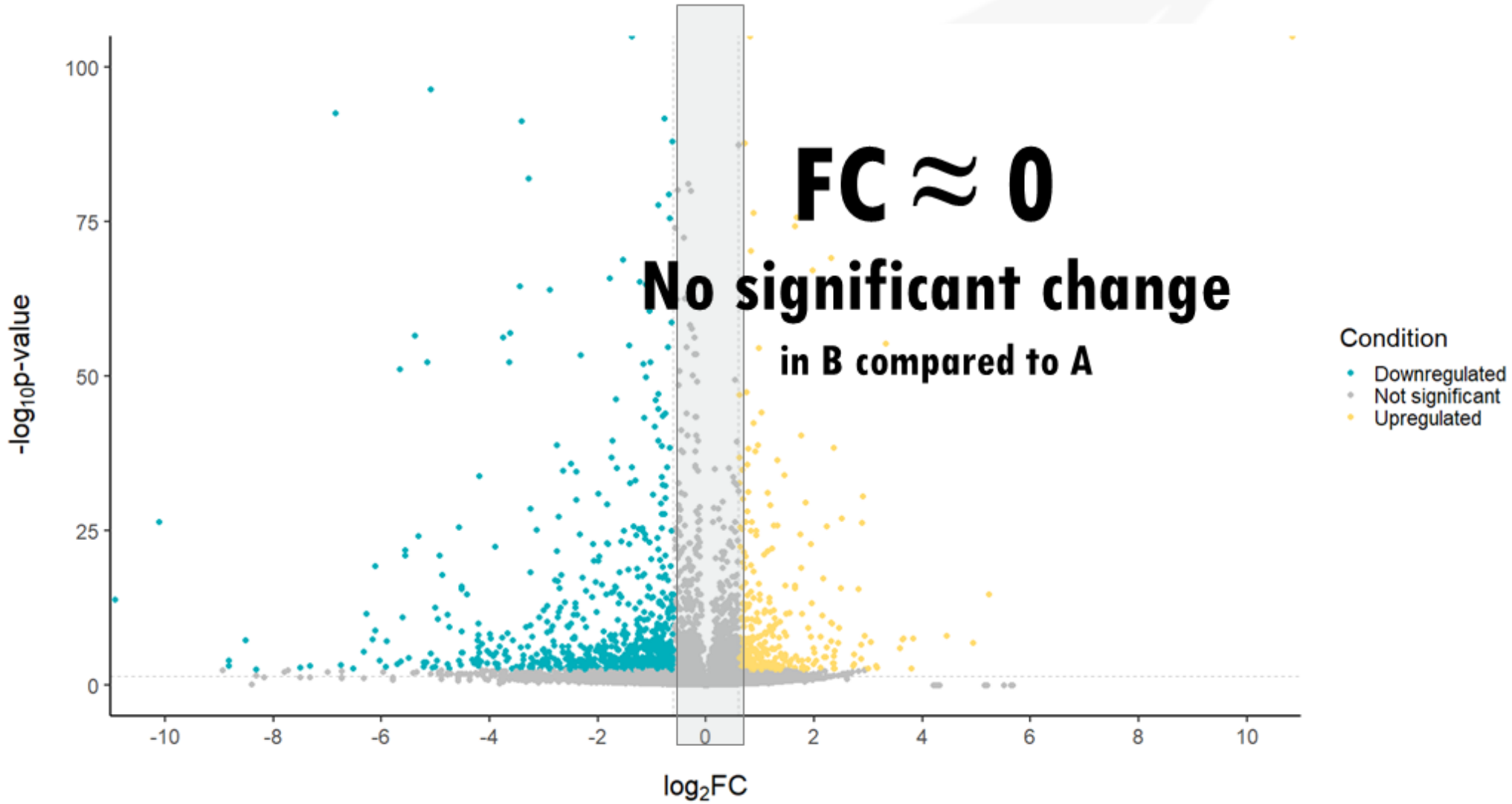
$$FC = \frac{\text{expression of gene X in B}}{\text{expression of gene X in A}}$$



$$FC = \frac{\text{expression of gene X in B}}{\text{expression of gene X in A}}$$

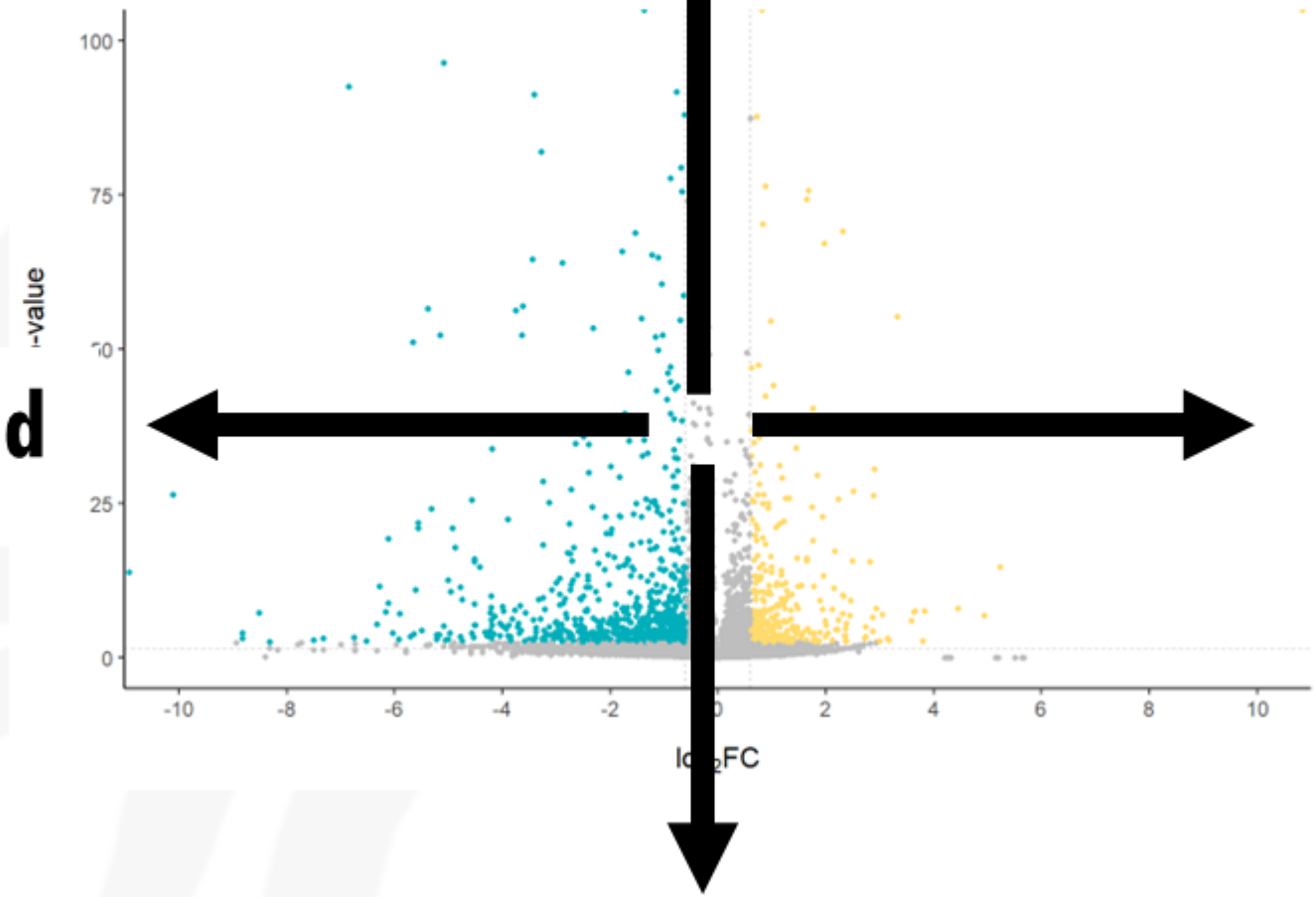


$$FC = \frac{\text{expression of gene X in } \mathbf{B}}{\text{expression of gene X in } \mathbf{A}}$$



Highly significant

Downregulated



Upregulated

Not significant

R Practical — DESeq2 pipeline

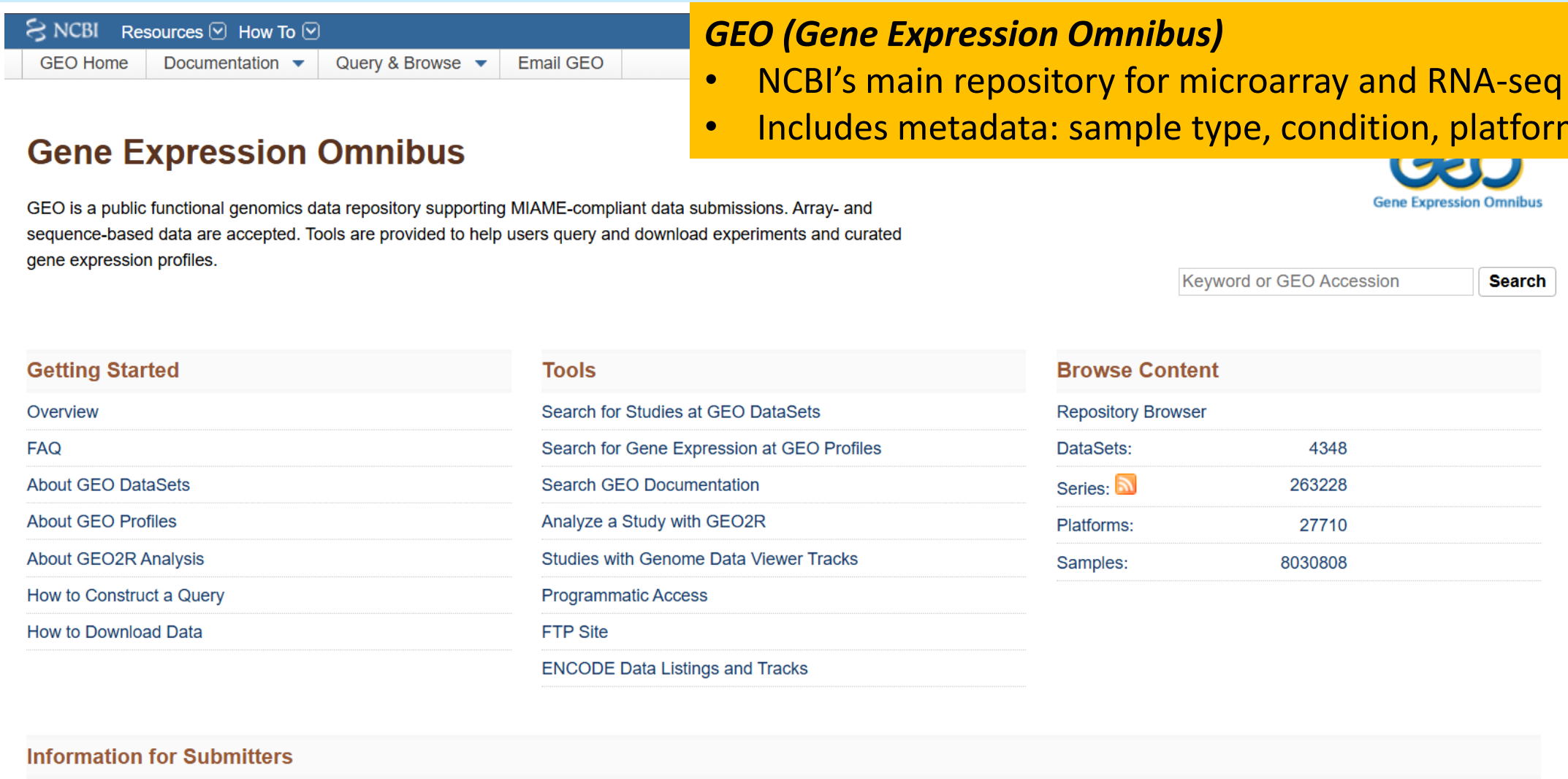
Data repositories, Pre-processing transcriptomic data and Visualisation

Public Repositories

Transcriptomics Data Repositories

GEO (Gene Expression Omnibus)

- NCBI's main repository for microarray and RNA-seq data
- Includes metadata: sample type, condition, platform



Install Packages






```r
# ============================================================
#  RNA-seq Analysis - CTLA4 Blockade with Ipilimumab
#  Dataset: GSE114716 (GEO)
#  CD4+ T cells - Melanoma patients
#  Baseline vs. Post-Ipilimumab (3 doses)
#  4 patients x 2 timepoints = 8 samples
#
#  Master's Course: Transcriptomics of RNA-seq
#  Verified pipeline - run step by step in class
# ============================================================

# 1. Install packages (run once)

install.packages("BiocManager")

install.packages(c("readxl", "ggplot2", "ggrepel"))
BiocManager::install(c("DESeq2", "GEOquery"))

# OR

install.packages("readxl")
install.packages("ggplot2")
install.packages("ggrepel")
BiocManager::install("DESeq2")
BiocManager::install("GEOquery")

```

Load Packages

```
27
28  # 2. Load packages
29  library(DESeq2)
30  library(GEOquery)
31  library(readxl)
32  library(ggplot2)
33  library(ggrepel)
34
```

Download GEO Dataset

```
35
36  # 3. Download GEO metadata and raw counts
37  gse_id <- "GSE114716"
38
39  gse <- getGEO(gse_id, GSEMatrix = TRUE)
40  meta <- pData(gse[[1]])
41
42  getGEOSuppFiles(gse_id, makeDirectory = TRUE)
43
44  counts_raw <- read_excel(
45    "GSE114716/GSE114716_raw.counts.hs.xlsx"
46  )
47
```

Prepare the count matrix

```r
# 4. Prepare count matrix
counts <- as.matrix(counts_raw[, -1])
rownames(counts) <- make.unique(as.character(counts_raw[[1]]))

storage.mode(counts) <- "numeric"
counts <- round(counts)
storage.mode(counts) <- "integer"

# Make sure sample IDs match the metadata
colnames(counts) <- as.character(names(counts_raw)[-1])

meta$description <- as.character(meta$description)
meta <- meta[match(colnames(counts), meta$description), ]

stopifnot(
  !anyNA(meta$description),
  identical(colnames(counts), meta$description)
)

```

Define contrasts

```
68
69  # 5. Define conditions and paired patients
70  meta$condition <- ifelse(
71    grepl("baseline", meta$title, ignore.case = TRUE),
72    "Baseline", "Post_Ipi"
73  )
74
75  meta$condition <- factor(
76    meta$condition,
77    levels = c("Baseline", "Post_Ipi")
78  )
79
80  # Samples 1&2, 3&4, 5&6, 7&8 are paired
81  meta$patient <- factor(
82    ceiling(as.numeric(meta$description) / 2)
83  )
84
85  rownames(meta) <- meta$description
86
87  stopifnot(
88    all(table(meta$patient, meta$condition) == 1)
89  )
90
```

Build DESeq2 dataset

```
84
85  # 6. Build DESeq2 dataset
86  dds <- DESeqDataSetFromMatrix(
87    countData = counts,
88    colData = meta,
89    design = ~ patient + condition
90  )
91
92  # Filter low-count genes
93  dds <- dds[rowSums(counts(dds) >= 10) >= 4, ]
94
95  # Run differential expression analysis
96  dds <- DESeq(dds)
97
```

Extract Results

```
 99  # 7. Extract results
100  res <- results(
101    dds,
102    contrast = c("condition", "Post_Ipi", "Baseline"),
103    alpha = 0.05
104  )
105
106  res_df <- as.data.frame(res)
107  res_df$gene <- rownames(res_df)
108
109  # Remove genes without adjusted p-values for plotting
110  plot_df <- res_df[!is.na(res_df$padj), ]
111
112  # Significance thresholds
113  plot_df$group <- "Not significant"
114
115  plot_df$group[
116    plot_df$padj < 0.05 &
117      plot_df$log2FoldChange >= 1
118  ] <- "Upregulated"
119
120  plot_df$group[
121    plot_df$padj < 0.05 &
122      plot_df$log2FoldChange <= -1
123  ] <- "Downregulated"
124
125
```

Select top 15 genes

```
126  # 8. Select top 15 upregulated and top 15 downregulated genes
127  up <- plot_df[plot_df$group == "Upregulated", ]
128  up <- head(up[order(up$padj), ], 15)
129
130  down <- plot_df[plot_df$group == "Downregulated", ]
131  down <- head(down[order(down$padj), ], 15)
132
133  top_genes <- rbind(up, down)
134
135  top_genes$negLogP <- -log10(pmax(top_genes$padj, 1e-300))
136
137  cat("Upregulated genes selected:", nrow(up), "\n")
138  cat("Downregulated genes selected:", nrow(down), "\n")
139
140  print(top_genes[, c("gene", "log2FoldChange", "padj")])
141
```

Volcano Plot with ggplot2

```r
# 9. Volcano plot
plot_df$negLogP <- -log10(pmax(plot_df$padj, 1e-300))

ggplot(plot_df, aes(x = log2FoldChange, y = negLogP)) +
  geom_point(aes(color = group), alpha = 0.7, size = 1.5) +
  geom_text_repel(
    data = top_genes,
    aes(label = gene),
    size = 3,
    max.overlaps = Inf
  ) +
  scale_color_manual(values = c(
    "Not significant" = "grey70",
    "Upregulated" = "red",
    "Downregulated" = "blue"
  )) +
  geom_vline(xintercept = c(-1, 1), linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), linetype = "dashed") +
  labs(
    title = "RNA-seq: Post-Ipilimumab vs Baseline",
    subtitle = "GSE114716 | DESeq2 | FDR < 0.05 and |log2FC| >= 1",
    x = "Log2 fold change",
    y = "-Log10 adjusted p-value",
    color = "Expression"
  ) +
  theme_classic()

# Save volcano plot
ggsave(
  "Volcano_DESeq2_GSE114716.pdf",
  width = 10,
  height = 8
)
```

Visualisation of results by Volcano Plot

